



WIESŁAW PIETRUSZKIEWICZ

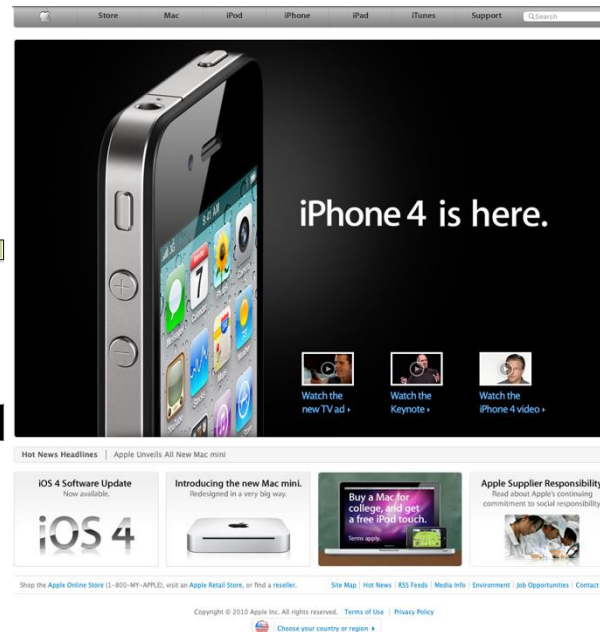
WEB DESIGN

Web Design Examples

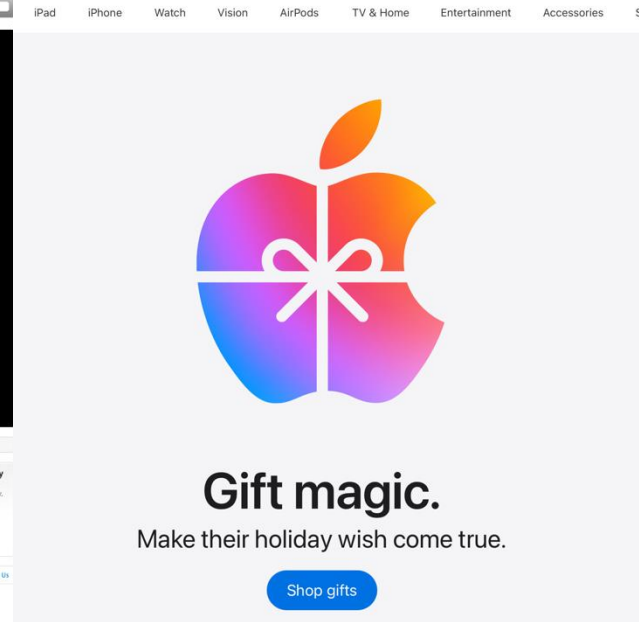
1998



2010



2024



Web Design Changes in Time

Standards

- Industry proposed & accepted
- Increased coverage

Higher bandwidth

- More multimedia
- HQ streaming

Better computers

- Increased web complexity
- Increased resolutions

Mobile devices

- Usage ad hoc
- Common availability

Users behaviour

- Increased interactivity
- Shorter attention span

The essence of web design

Web design is essential to company's web presence (especially e-commerce - the main source of its founding) for several reasons, as it directly impacts user experience, brand perception, and ultimately, sales!

Importance of web design #1

First Impressions

- A well-designed website creates a positive first impression. Users often form opinions about a brand within seconds of visiting a site. A professional and visually appealing design can instil trust and encourage visitors to explore further.

User Experience (UX)

- Effective web design focuses on user experience, ensuring that navigation is intuitive and that users can easily find what they are looking for. A seamless UX reduces frustration and increases the likelihood of conversions.

Mobile Responsiveness

- With a significant portion of online shopping occurring on mobile devices, responsive web design is crucial. A site that adapts to different screen sizes ensures that users have a consistent and enjoyable experience, regardless of the device they use.

Brand Identity

- Web design is a key component of a brand's identity. Consistent use of colours, fonts, and imagery helps reinforce brand recognition and loyalty. A strong brand presence can differentiate a business from its competitors.

Loading Speed

- Design elements can affect website loading speed. A well-optimized site with efficient design can load quickly, which is critical for retaining visitors. Slow-loading sites can lead to high bounce rates and lost sales.

Importance of web design #2

Call-to-Action (CTA) Placement

- Effective web design strategically places CTAs to guide users toward desired actions, such as making a purchase or signing up for a newsletter. Clear and compelling CTAs can significantly improve conversion rates.

SEO Optimization

- Good web design incorporates SEO best practices, making it easier for search engines to crawl and index the site. This can improve visibility in search results, driving more organic traffic to the e-commerce site.

Trust and Credibility

- Elements such as customer reviews, security badges, and professional design contribute to a sense of trust and credibility. Users are more likely to make purchases from sites that appear trustworthy and secure.

Accessibility

- Inclusive web design ensures that all users, including those with disabilities, can access and navigate the site. This not only broadens the potential customer base but also complies with legal standards in many regions.

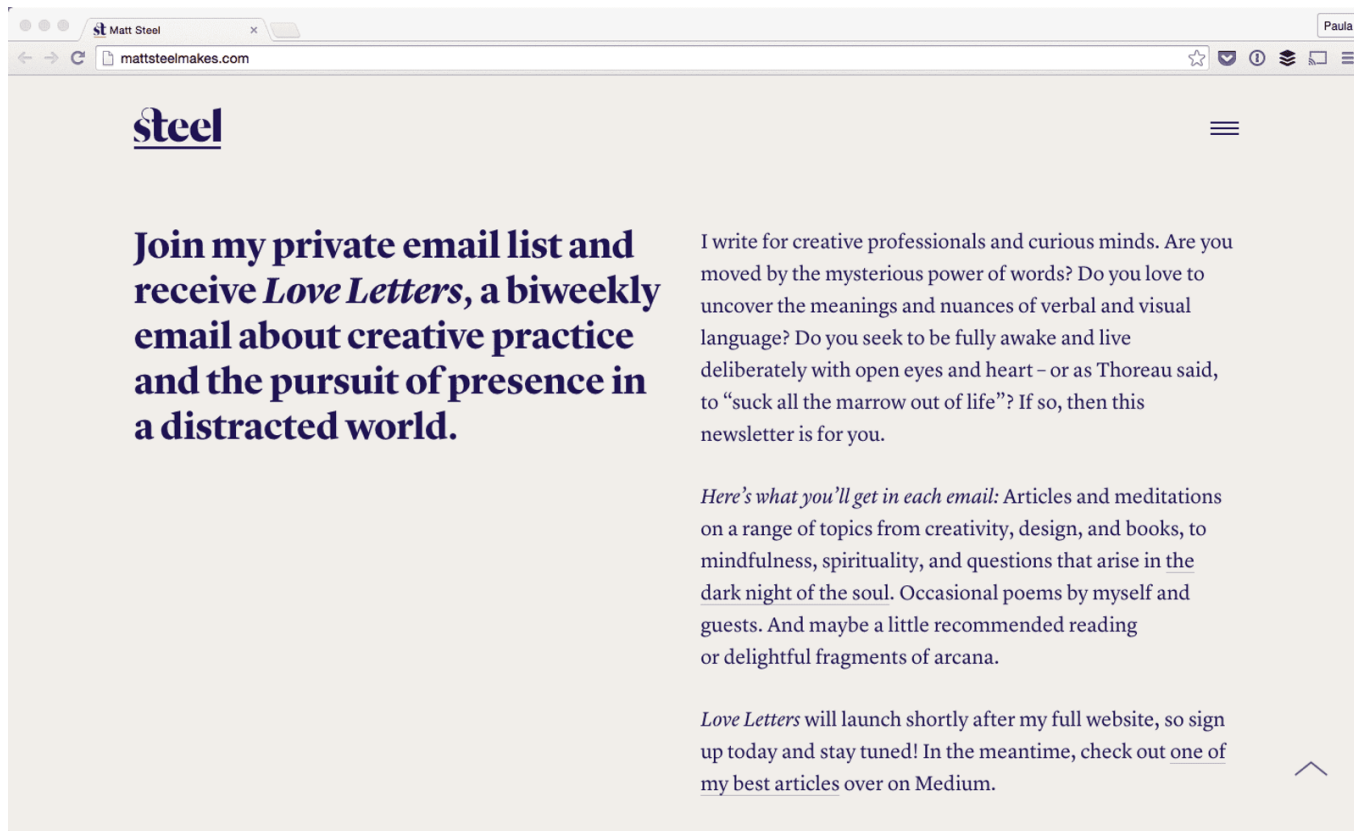
Analytics and Feedback

- A well-designed e-commerce site can integrate analytics tools that help track user behaviour. This data can inform design improvements and marketing strategies, leading to better performance over time.

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

1. Use readable and web-friendly fonts



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

2. Utilise the 'F' pattern



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

3. Or the 'Z' pattern



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

4. Use negative space



iPhone X 

**The ultimate iPhone.
In two sizes.**

Starting at A\$1,629

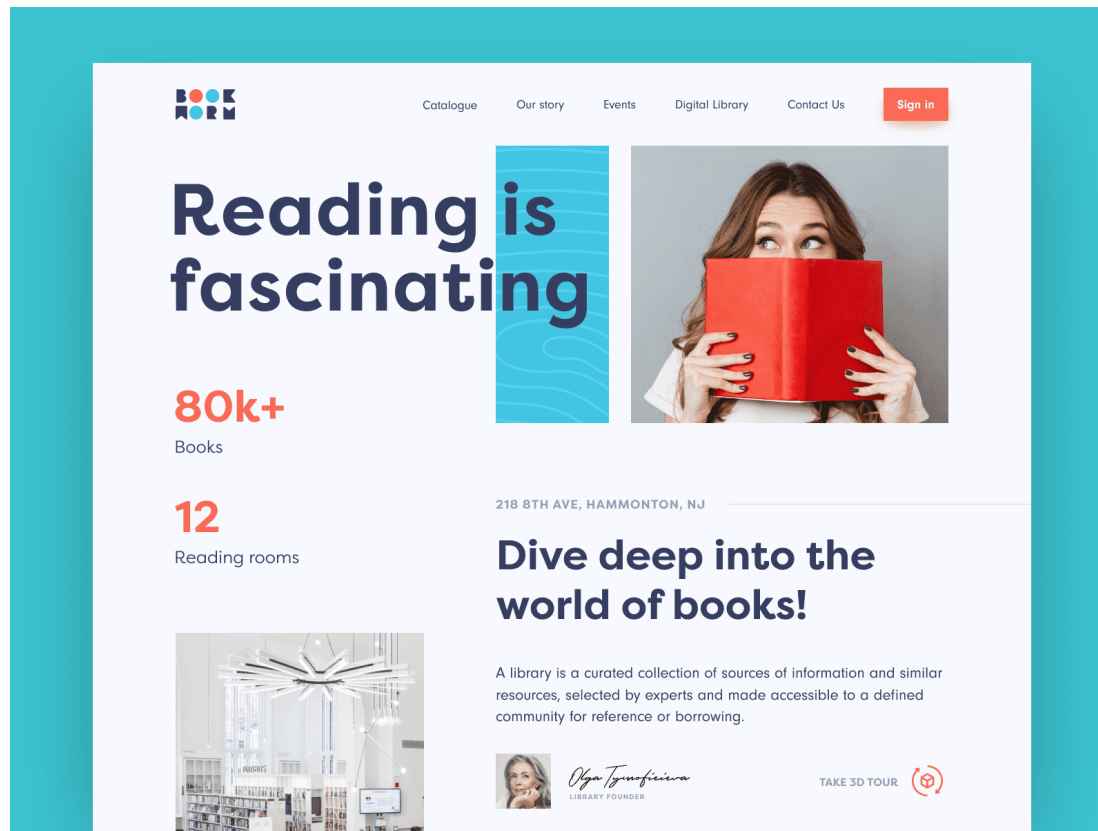
[Buy](#)

[Learn more >](#)

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

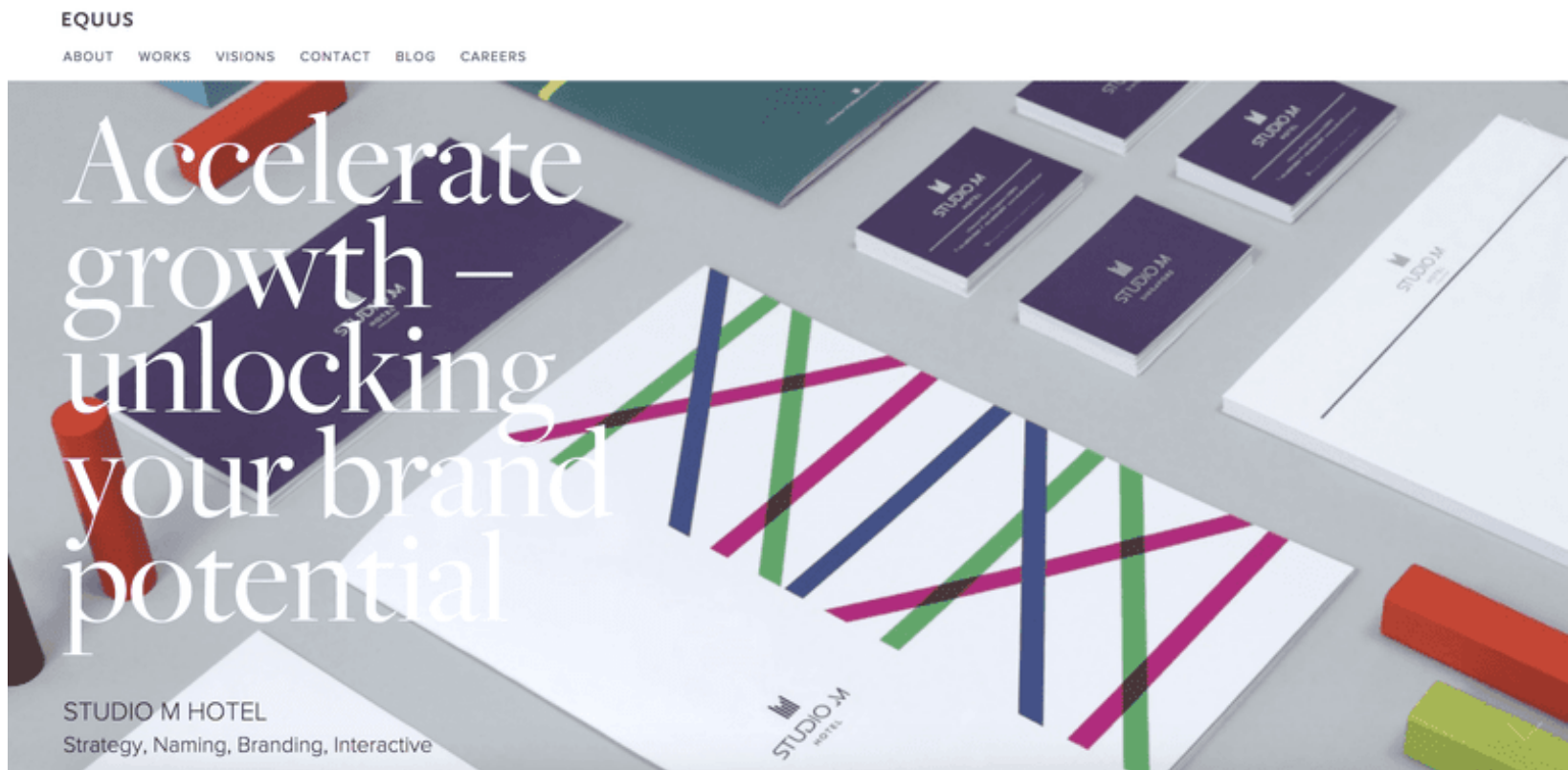
5. Keep your design consistent



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

6. Simple and logical page navigation



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

7. Use a complementary colour palette



Complementary colours

Encyclopaedia Britannica

Color wheel



RYB color model

© Encyclopædia Britannica, Inc.

Colour perception

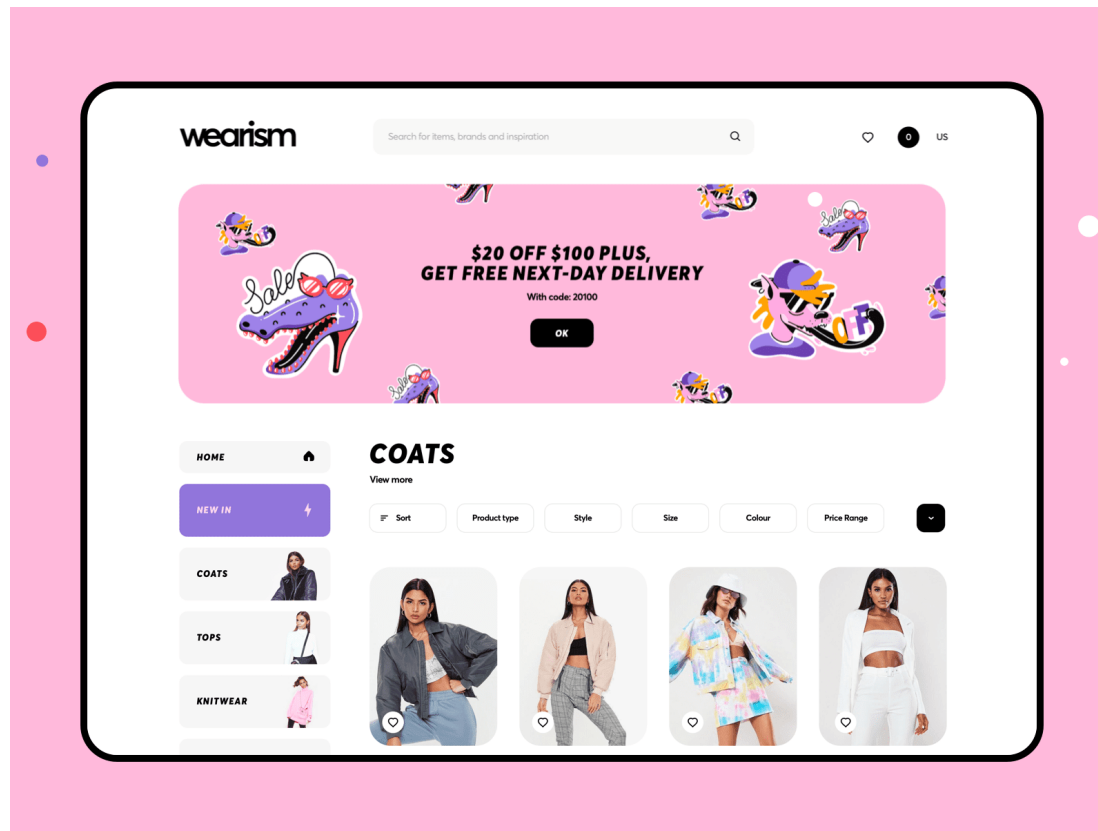
London Image Institute (londonimageinstitute.com)

Red Excitement Strength Love Energy	Orange Confidence Success Bravery Sociability	Yellow Creativity Happiness Warmth Cheer	Green Nature Healing Freshness Quality	Blue Trust Peace Loyalty Competence
Pink Compassion Sincerity Sophistication Sweet	Purple Royalty Luxury Spirituality Ambition	Brown Dependable Rugged Trustworthy Simple	Black Formality Dramatic Sophistication Security	White Clean Simplicity Innocence Honest

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

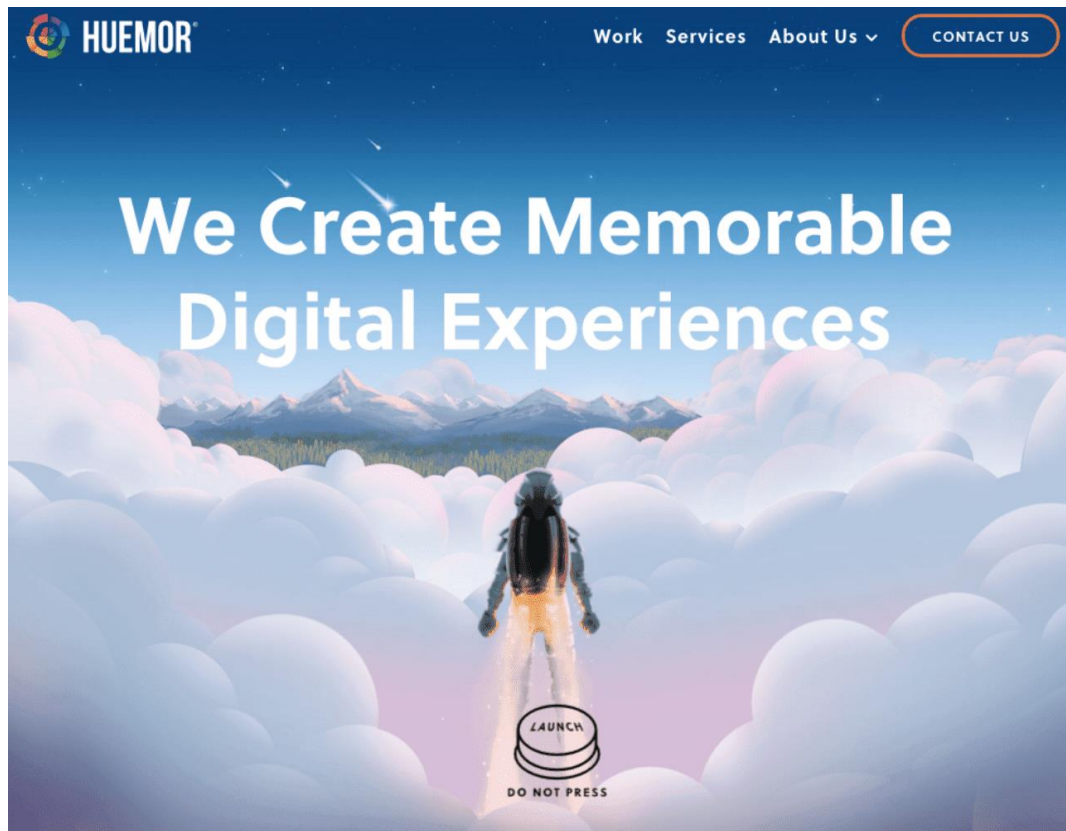
8. Keep the audience in mind



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

9. Optimize buttons and calls-to-action



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

10. Maintain a visual hierarchy



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

11. Pay attention to the details



FREE SHIPPING — on all orders over \$35*



CARBON
BEAUTY

PRIVACY POLICY
TERMS & CONDITIONS
ABOUT



SHIPPING INFO
RETURNS / EXCHANGES
CONTACT

WEEKLY NEWSLETTER

NAME@EMAIL.COM

SUBSCRIBE

© 2019

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

12. Use Fitts's Law

Read micro-articles about copywriting, content marketing, and psychology [HERE](#). ×



[WELCOME](#) [ABOUT](#) [SERVICES](#) [ARTICLES](#) [BOOKS](#) [CONTACT](#)

LEARN TO PERSUADE

Micro-articles about copywriting, content marketing, and psychology.

COMPLETE LIBRARY

Fitts's law #1

Fitts's Law is a predictive model of human movement that describes the relationship between the distance to a target and the size of the target, and how these factors affect the time it takes to reach the target. Formulated by psychologist Paul Fitts in 1954, the law is often used in the fields of ergonomics, human-computer interaction, and design.

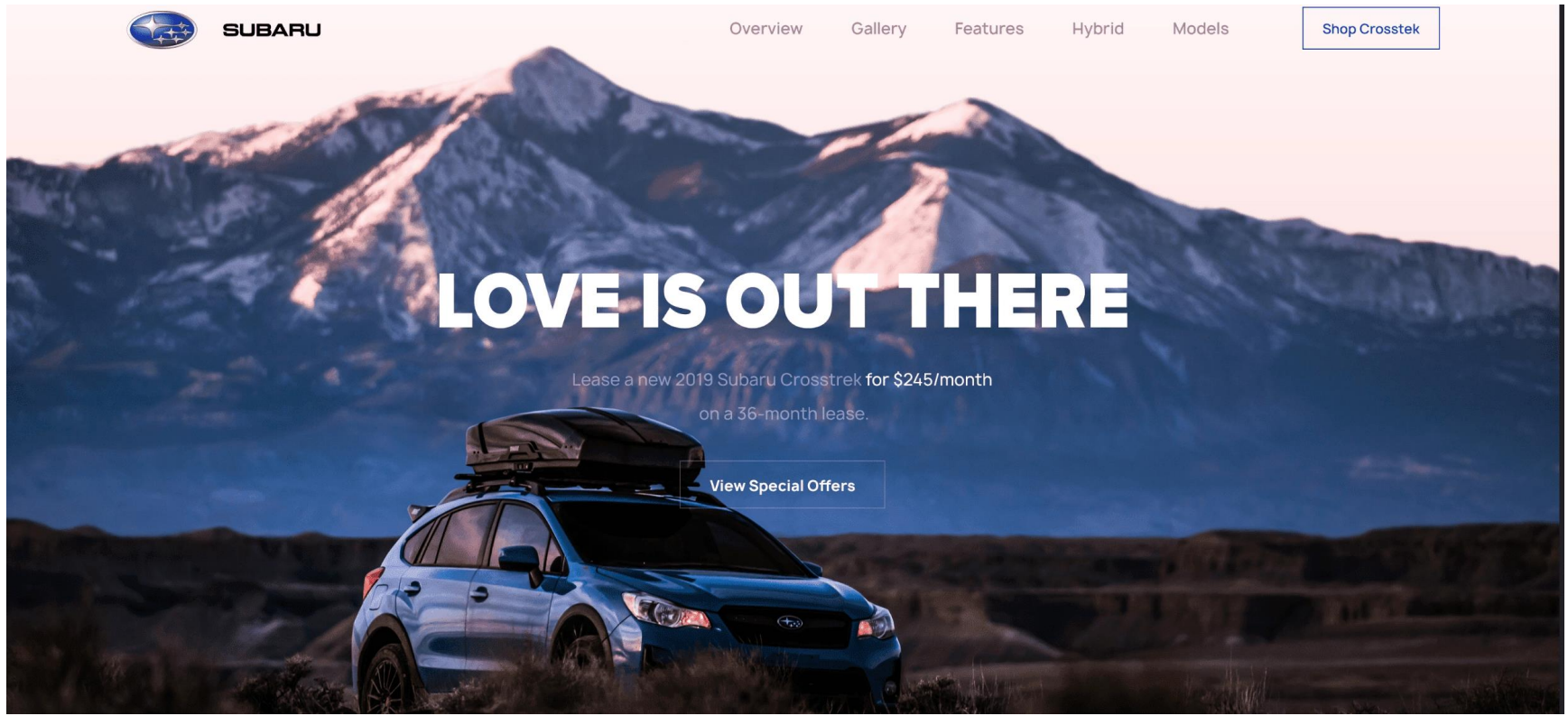
Fitts's law #2

- **Distance and Size:** The farther away a target is, and the smaller it is, the longer it will take to reach it. This means that larger and closer targets are easier and quicker to hit.
- **Logarithmic Relationship:** The relationship is logarithmic, meaning that as the distance increases or the size decreases, the increase in time is not linear but rather diminishes at a decreasing rate.
- **Applications:** Fitts's Law is widely applied in user interface design, where it informs the placement and sizing of buttons and controls to optimize user interaction. For example, larger buttons that are closer to the user's cursor can improve efficiency and reduce errors.

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

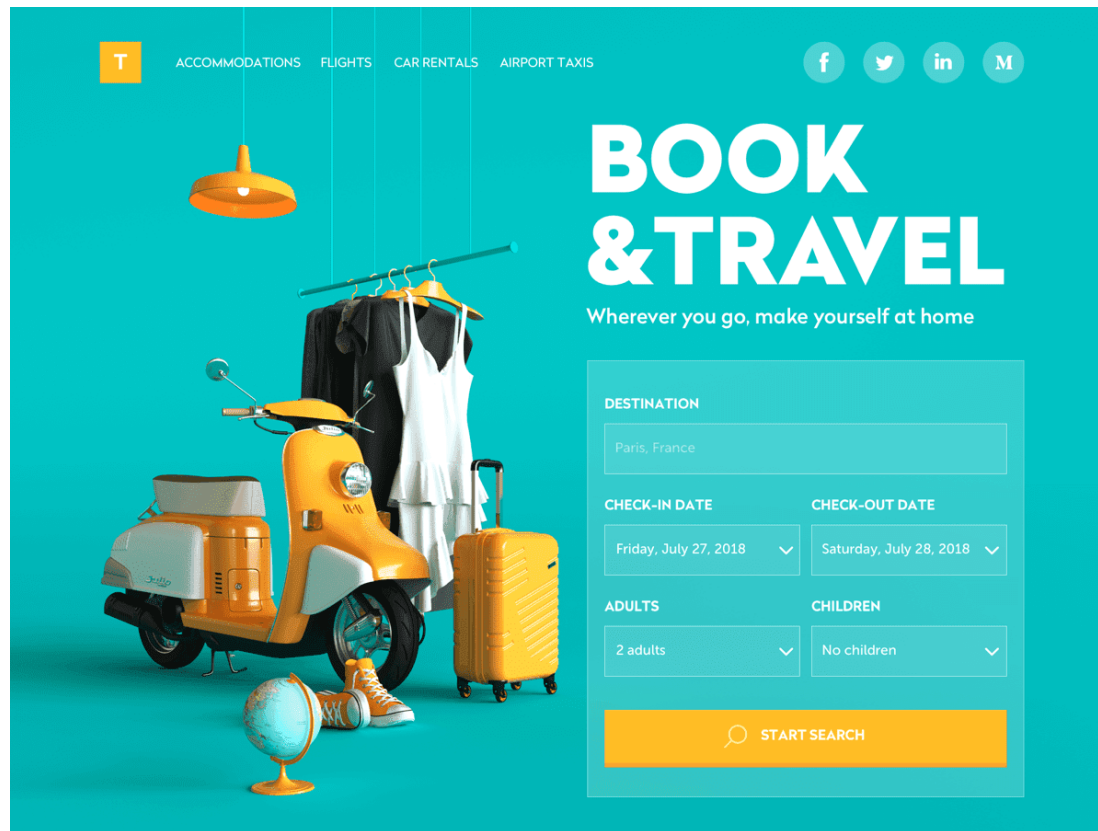
13. Choose your images wisely



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

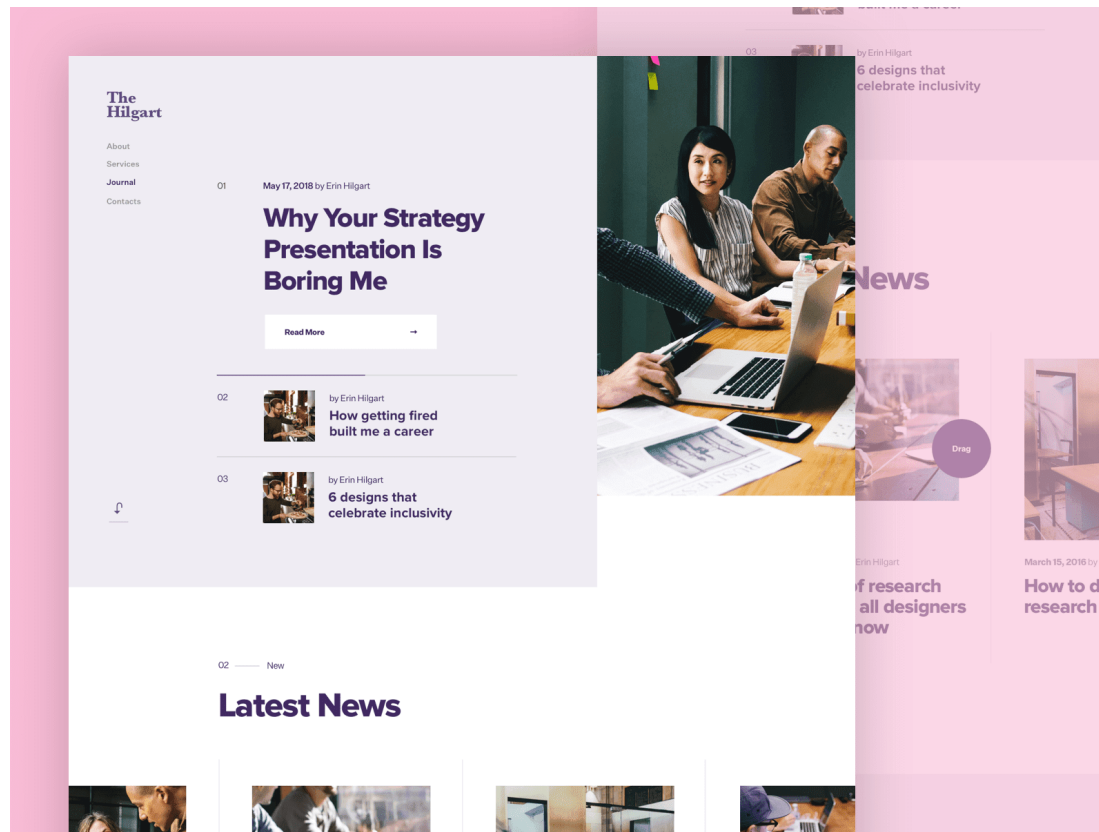
14. Prioritize the user experience



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

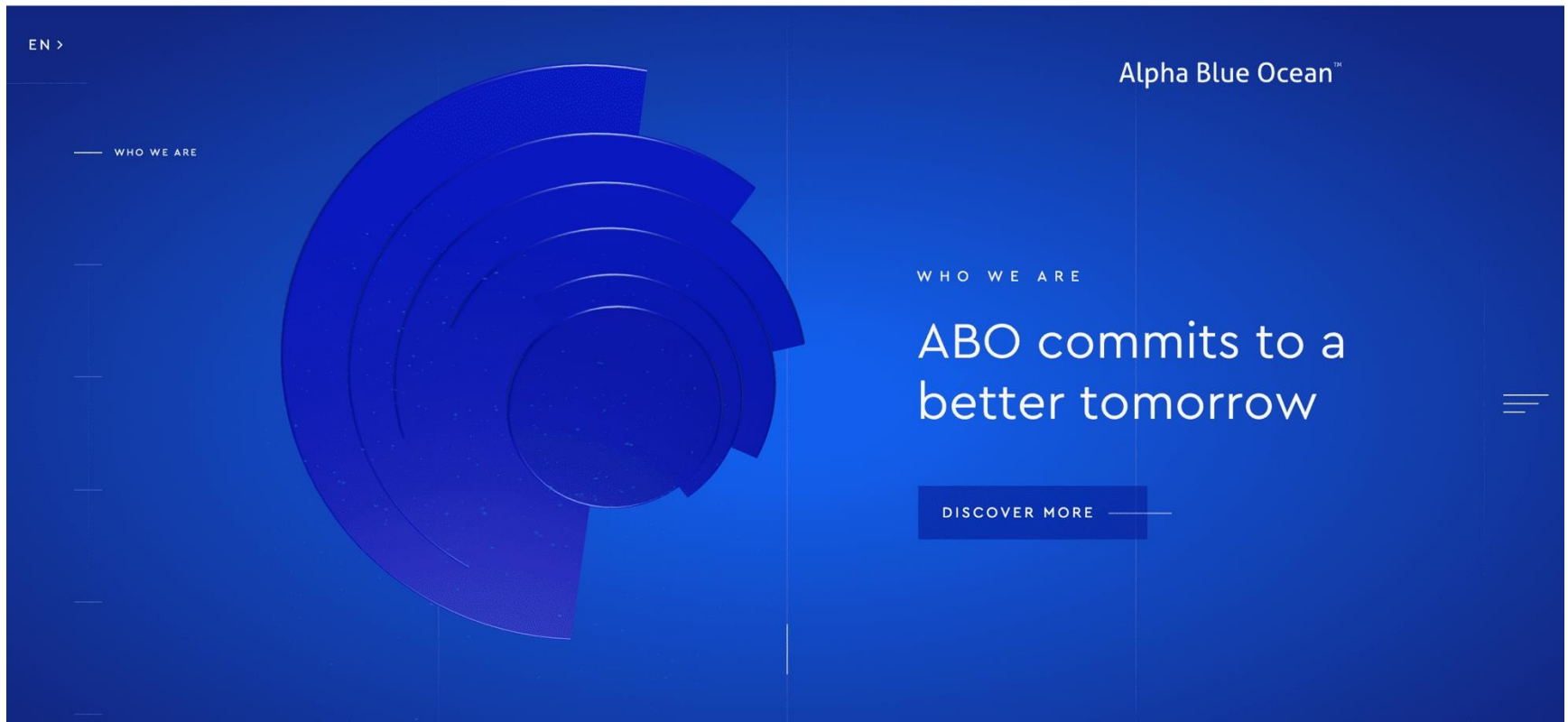
15. Consider using grid systems



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

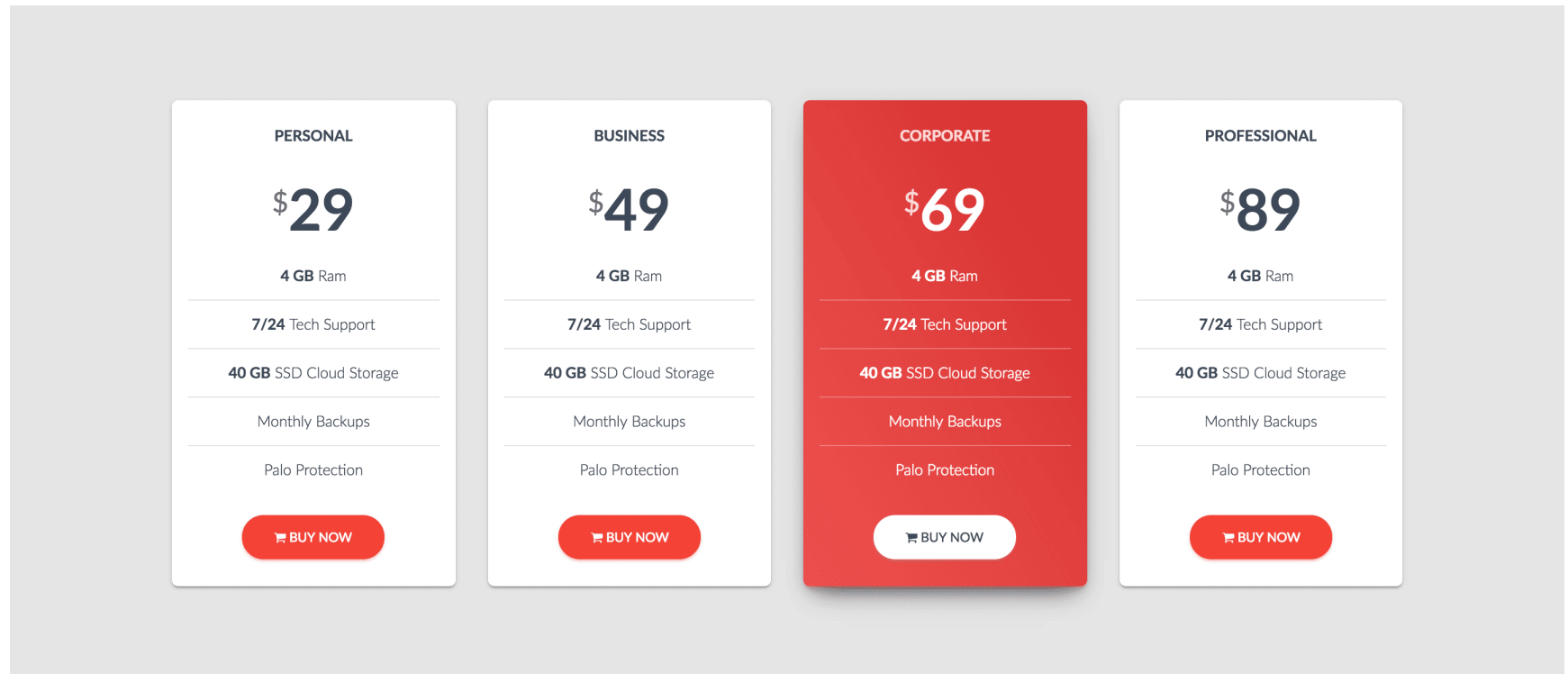
16. Avoid big chunks of text



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

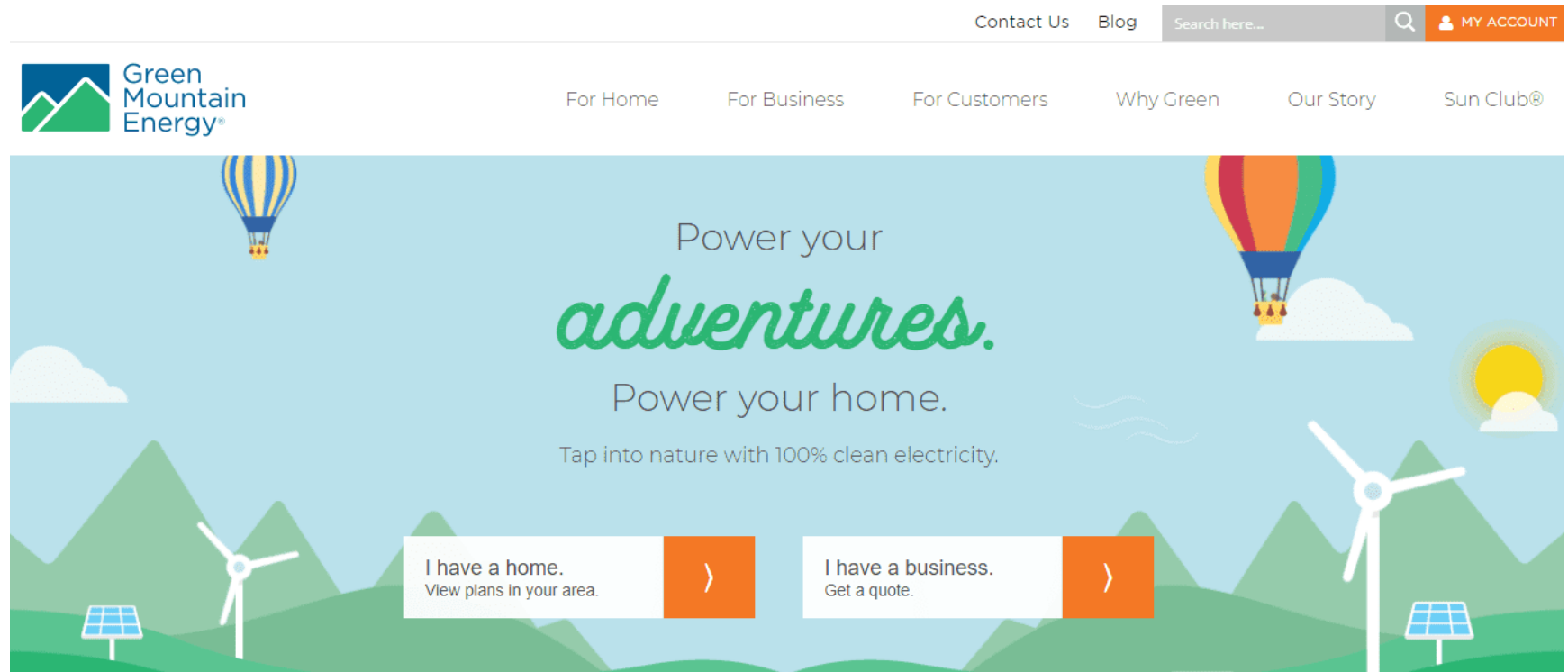
17. Use invariance



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

18. Hick's Law



Hick's law #1

Hick's Law, formulated by British psychologist William Edmund Hick in the 1950s, describes the relationship between the number of choices available and the time it takes for an individual to make a decision. The law states that the more options a person has, the longer it will take them to make a choice.

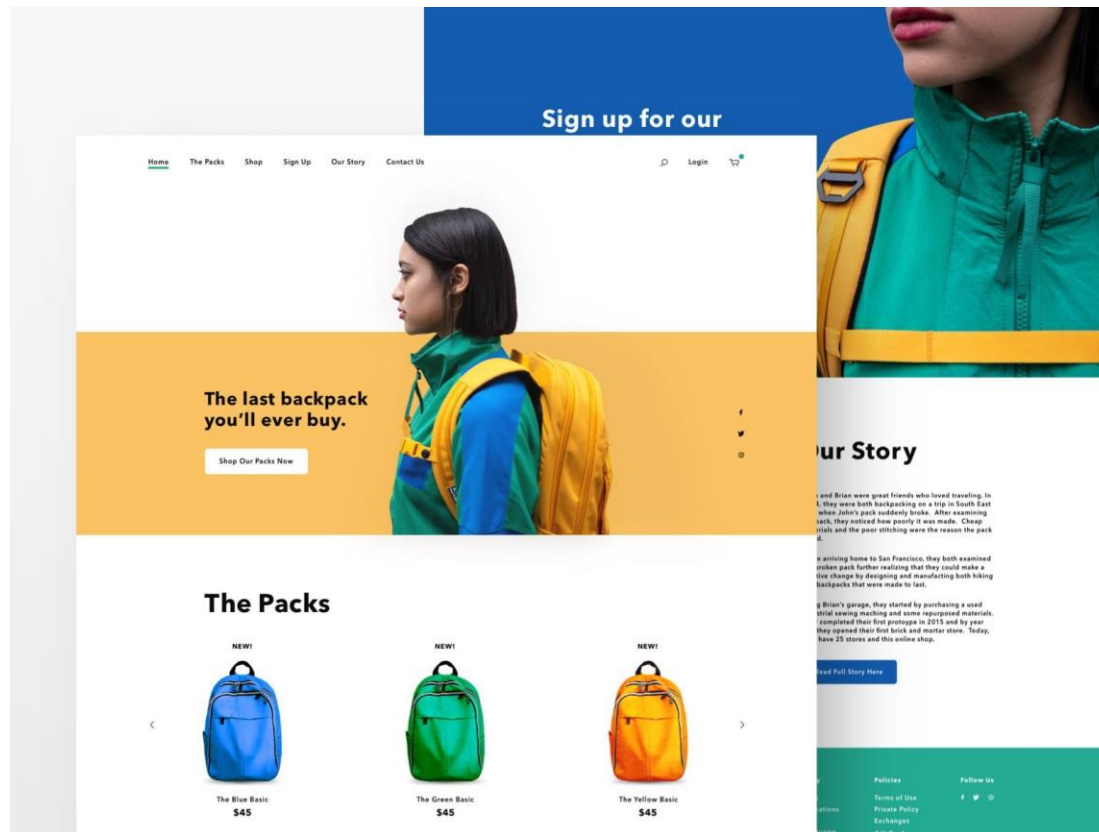
Hick's law #2

- **Choice Complexity:** As the number of options increases, the decision-making time increases logarithmically. This means that while adding more choices does increase decision time, the effect diminishes as the number of options grows.
- **Cognitive Load:** More choices can lead to increased cognitive load, which may result in decision fatigue or choice paralysis, where individuals struggle to make a decision due to overwhelming options.
- **Applications:** Hick's Law is widely applied in fields such as user interface design, marketing, and behavioural economics. It suggests that simplifying choices can enhance user experience and improve decision-making efficiency.

20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

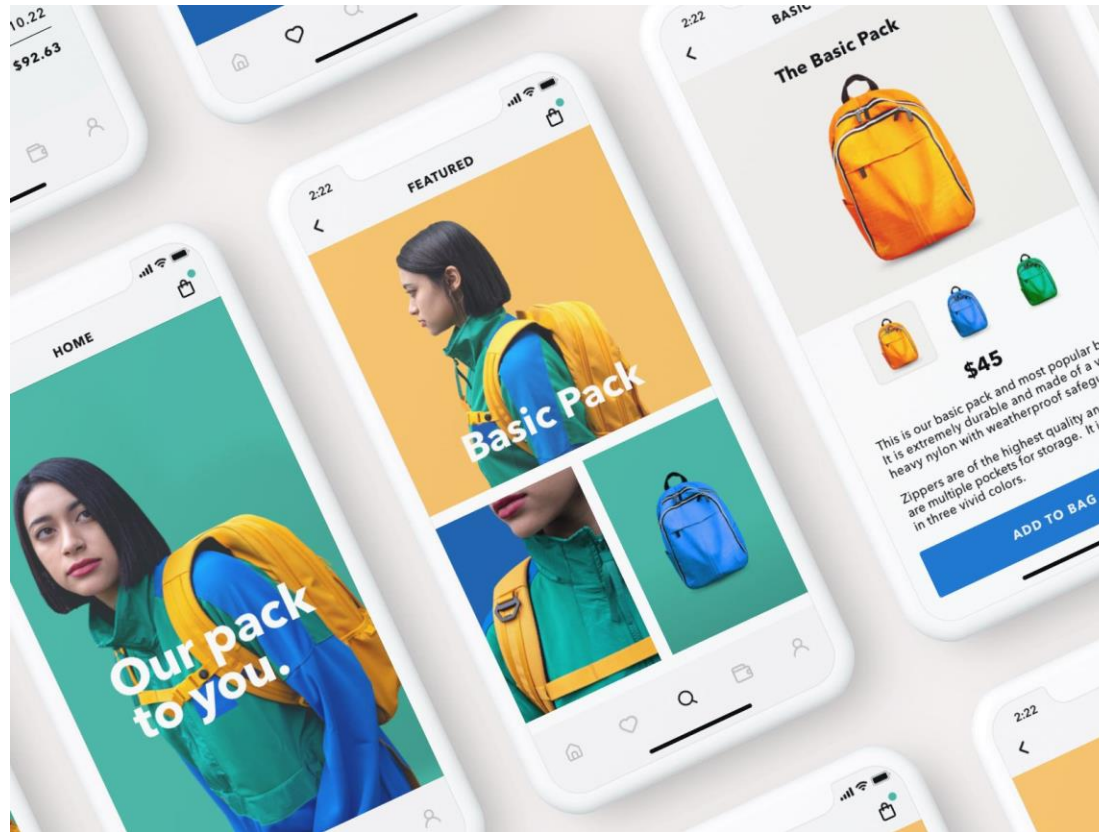
19. Use symmetry



20 Web Design Principles by Canva.com

<https://www.canva.com/learn/20-web-design-principles-follow/>

20. Design for both web and mobile



Website Layout

- The website layout defines placing of content (text, images, multimedia)
- For 10 Website Layouts see (following slides):
<https://websitesetup.org/website-layouts/>
- Compare with:
<https://xd.adobe.com/ideas/principles/web-design/11-website-layouts-that-made-content-shine-in-2019/>

10WL – Single Column Layout



10WL – Single Column Layout

- Simple
- Reusable across different devices (mobile, laptops, large displays)
- Focus on reading
- No distractions
- Longer layout (vertically), often requiring scrolling

10WL – Content Focused Layout

boagworks & boagworld



Calls to Action: 15 Techniques for Improvement, With Examples

AUTHOR:

[Paul Boag](#)

DATE:

6 June 2019

CATEGORY:

[Interface Design](#), [Digital Marketing](#)

READING TIME:

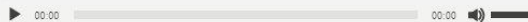
28 minutes

Like this email? Get them delivered to your inbox.

- Free Courses
- Free Books
- Exclusive Insights
- And Much More!

Where do you work? 

Every website should have a call to action, a response you want users to complete. But how do you encourage users to act? How do you create an effective call to action?



Listen to this post and subscribe using [Apple Podcasts](#), [Google Play](#), [Spotify](#) or [RSS](#)

Almost all websites have some form of a call to action, from signing up to a newsletter to submitting a contact us form. However, getting a user to complete a call to action is more complicated than one would first think. There are many factors at play that you need to consider.

The Factors That Influence Why People Complete a Call to Action

10WL – Content Focused Layout

- Primary column for content and side(s) columns for additional information
- Common for news sites or blogs
- Hint – content should be broken onto smaller chunks (headings, lists, quotes etc.)
- Side columns should not distract from the primary one

10WL – Magazine Layout

Q

Sections

The Washington Post
Democracy Dies in Darkness

Get one year for \$29 USD

Sign in

Coronavirus

LIVE UPDATES

Access to these updates is free

FDA panel to consider Pfizer vaccine, as U.S. reports 3,140 deaths in a day

- 7:50AM
Saudi Arabia approves Pfizer-BioNTech vaccine for use
- 7:25AM
Surge of cases in South Africa driven by partying youth, health minister says
- 6:55AM
White House task force recommends lifting ban on travelers from Europe, Brazil, report says

For some Americans, the coronavirus vaccine can't come soon enough. Others are taking a wait-and-see

Lessons from the pandemic fall: Infections are rare in classrooms, not off campus



Harry Brighouse, professor of philosophy at the University of Wisconsin at Madison, teaches an in-person class on Nov. 17. (Lauren Justice for The Post)

Virial testing, contact tracing, quarantine housing and creative public health campaigns have become the go-to tools of higher education.

By Nick Anderson and Susan Svrluga

Stealing to survive: More Americans are shoplifting food as aid runs out during the pandemic



Election 2020

LIVE UPDATES

Biden headed to Georgia next week to campaign for Senate candidates

- 7:32AM
Trump pressures congressional Republicans to help in his fight to overturn the election
- 6:54AM
Biden to campaign for Senate candidates in Georgia next week; Pence returns to the state today

Biden selects Katherine Tai as U.S. trade representative

Tai has been the chief trade counsel on the House Ways and Means Committee since 2017.

By Amy B Wang and David J. Lynch

Who Joe Biden is picking to fill his White House and Cabinet

Biden team explores Sen. Doug Jones for attorney general nominee as the candidate list narrows to

Opinions

If Trump is serious about running again, he should study Nixon

Opinion by T.R. Reid

Repeating 'listen to the science' won't undo the damage Trump has done

Opinion by E.J. Dionne Jr.

Latino voters are complex, but their problems are not. Here's how Biden can deliver solutions.

Opinion by Julio Ricardo Varela

Incarcerated people are suffering from covid-19 more than most. They should be among the first vaccinated.

Opinion by Danielle Allen

The danger is growing that


Web Design Principles

10WL – Magazine Layout

- Typical for magazines or news sites, containing a large number of different pieces of information
- Layout inspired by the traditional printing, uses combination of headlines and imagery
- Content provided for each news is only teasing
- Problematic for responsive webpages
- Very overwhelming visually

10WL – Grid Breaking Layout



10WL – Grid Breaking Layout

- Breaks webpage into a grid
- Allows to draw attention to particular screen elements
- More complex in management and design (responsiveness e.g. if overlapping was used)
- The flow of attention differs from the natural one

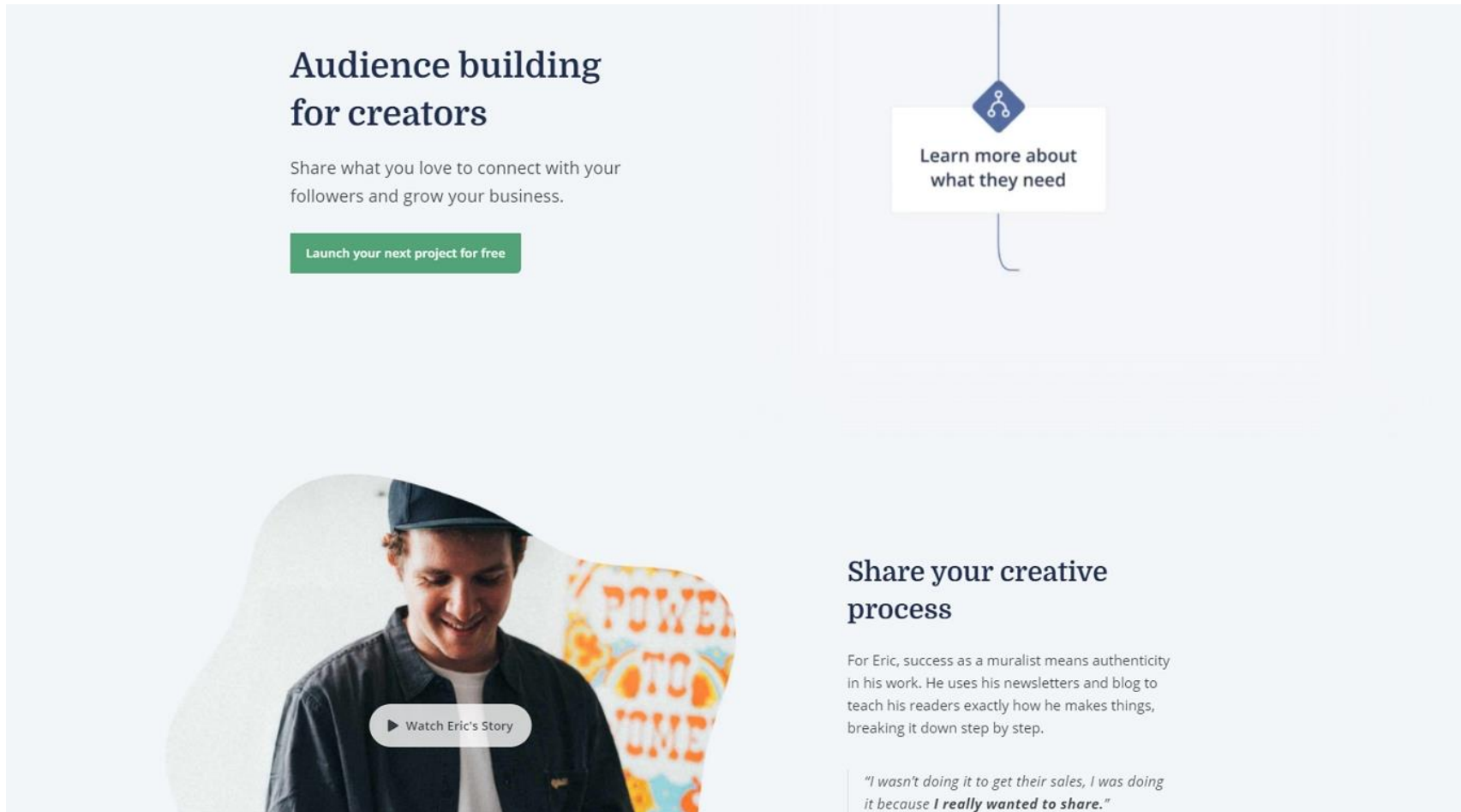
10WL – Full-Screen Layout



10WL – Full-Screen Layout

- Single screen, not requiring to scroll
- Storytelling or presentations
- Commonly uses multiple images or multimedia
- More difficult in management and design

10WL – Alternating Layout



10WL – Alternating Layout

- A series of content blocks
- Each block has two-column layout (usually image and text)
- The sides (image<->text) are changed between following blocks
- A good choice for webpages providing a mix of information (imagery with text)
- A simple layout

10WL – Card-Based Layouts



ASOS DESIGN Petite slouchy chino trouser in mushroom

€37.99



ASOS DESIGN Tall mid rise '90's' straight leg in chocolate PU

€47.99



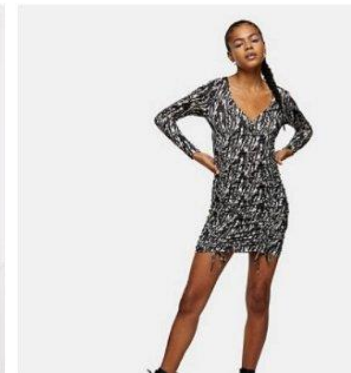
adidas Originals 'Relaxed Risqué' long sleeve top in beige

€34.95



Object twist from top in metallic snake print

€41.99



10WL – Card-Based Layouts

- Used to give user a possibility to choose between different options
- Each option with information (image and text)
- Typical choice for listing pages
- Easy to provide responsiveness (number of cards depends on the available width)
- Potential problems – a lack of image, different length of content

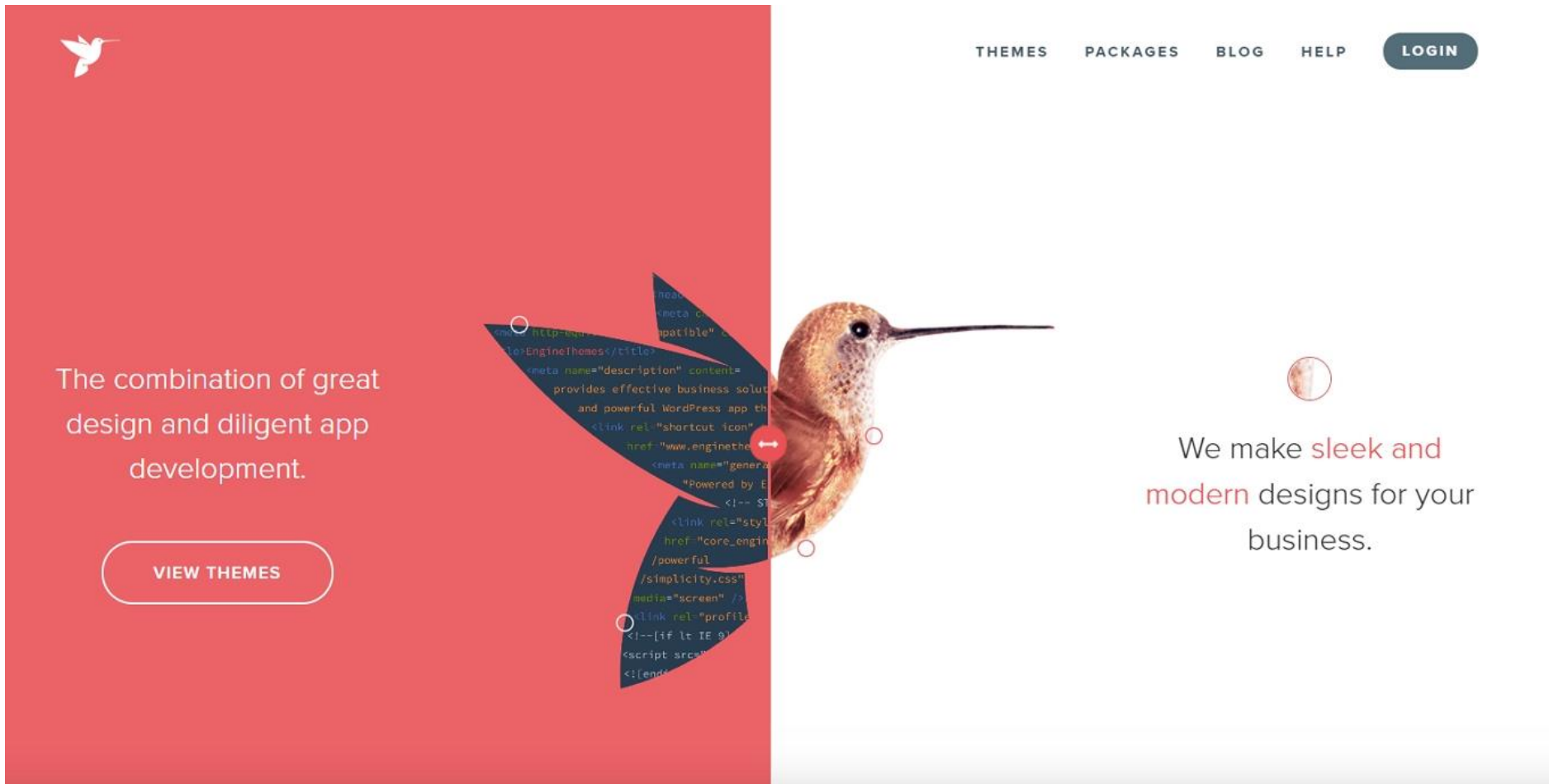
10WL – Hero Layout



10WL – Hero Layout

- Layout focuses on prominent products (services), providing their large banners on top
- Essential for value proposition
- Notable usage – Apple's new products
- Could grab the attention from other elements

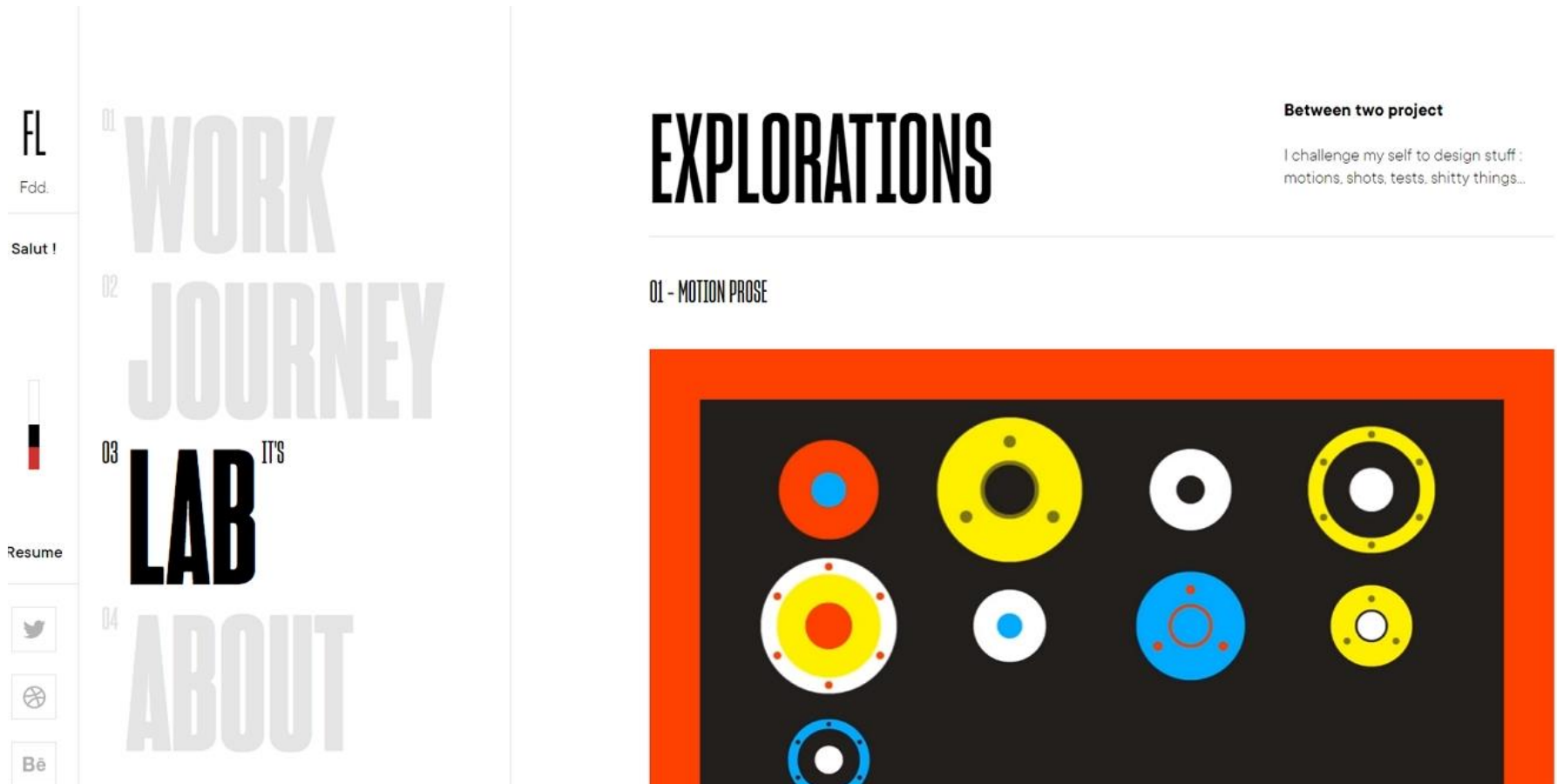
10WL – Split Screen Layout



10WL – Split Screen Layout

- A distinct separation of the screen onto two elements
- Very effective to provide a binary choice
- Usually symmetric split (nonsymmetric split could suggest the choice)
- A limiting option in other scenarios

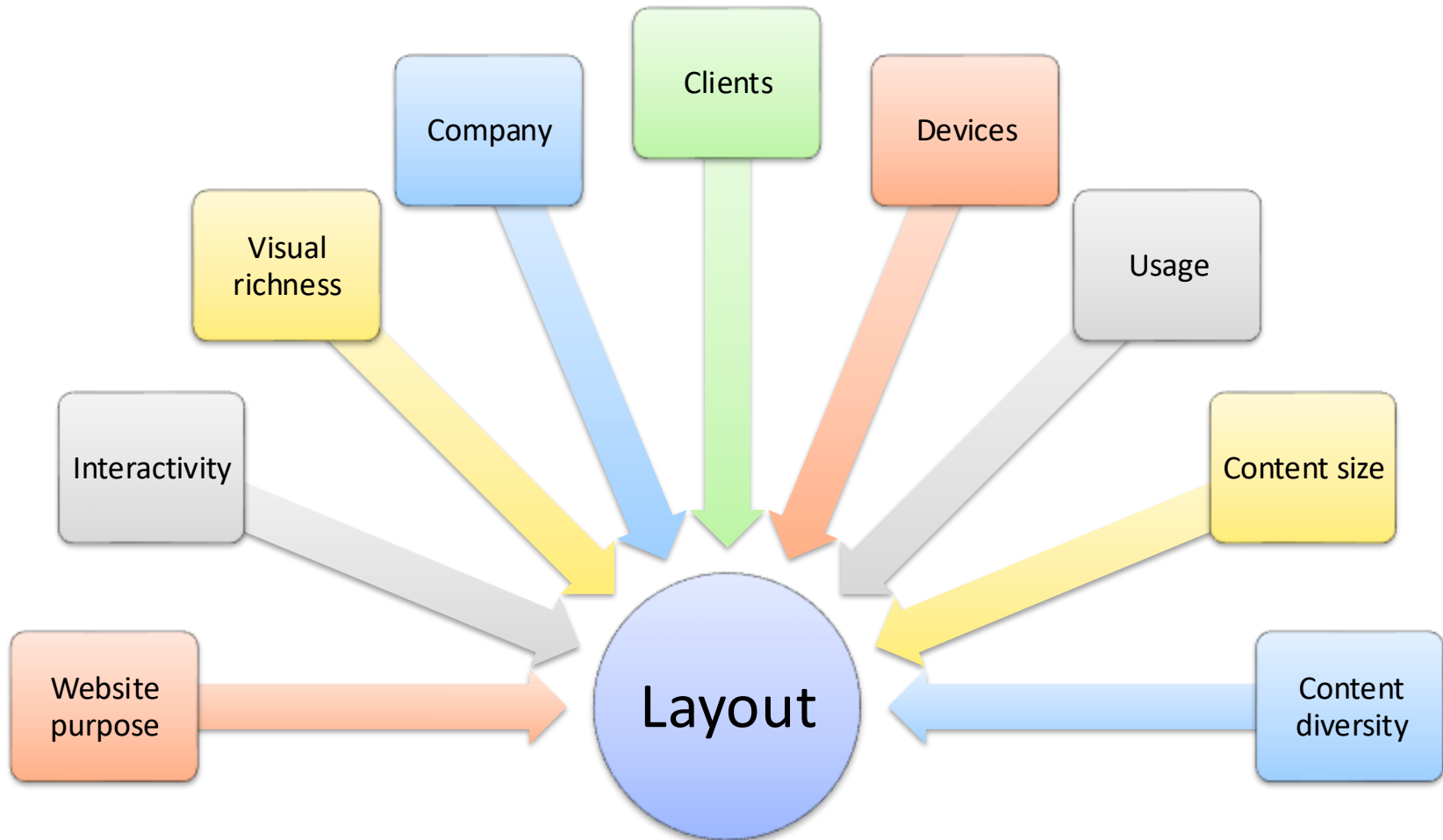
10WL – Asymmetrical Layout



10WL – Asymmetrical Layout

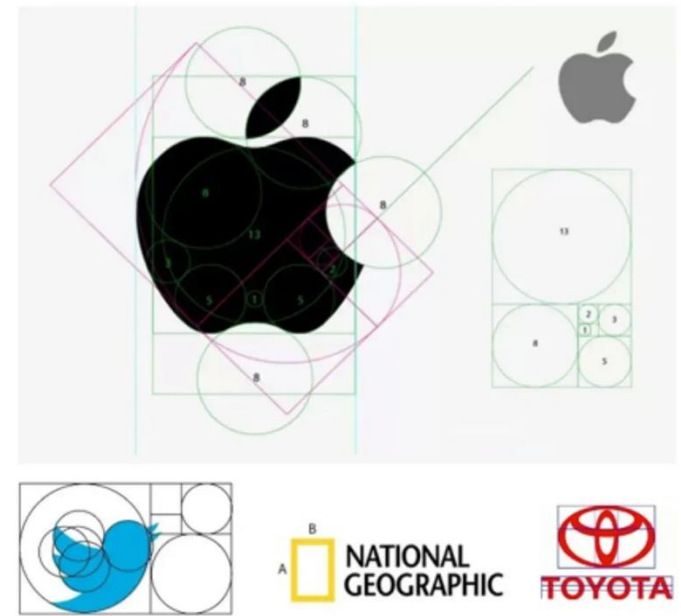
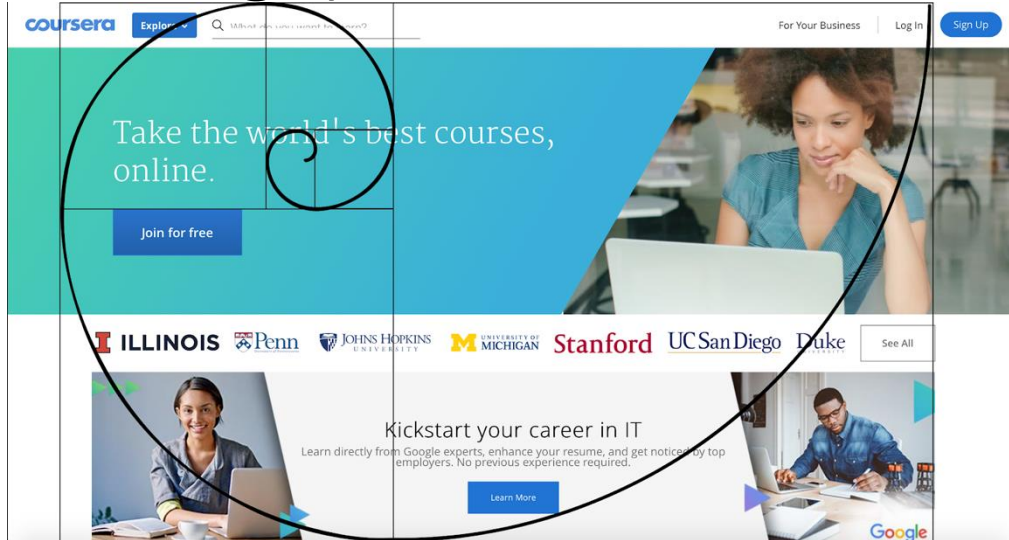
- Divides the viewport onto two different sides
- Asymmetry gives a chance to emphasise one of them
- Thus, it causes content to be divided on primary and secondary one
- Secondary containing e.g. less important details or navigation

Layout selection – Factors



The golden ratio

- Golden ratio: $\frac{a+b}{a} = \frac{a}{b} = \varphi = \frac{1+\sqrt{5}}{2} = 1.618$
- Commonly used in design (including web design)



The golden ratio in web design

- **Proportionality:** The Golden Ratio is based on the idea that certain proportions are more visually appealing. When elements are sized or spaced according to the Golden Ratio, they tend to create a sense of harmony and balance.
- **Layout and Grid Systems:** Designers can use the Golden Ratio to create grid systems for layouts. By dividing a page into sections that follow the Golden Ratio, designers can achieve a balanced and harmonious structure.

The golden ratio in web design

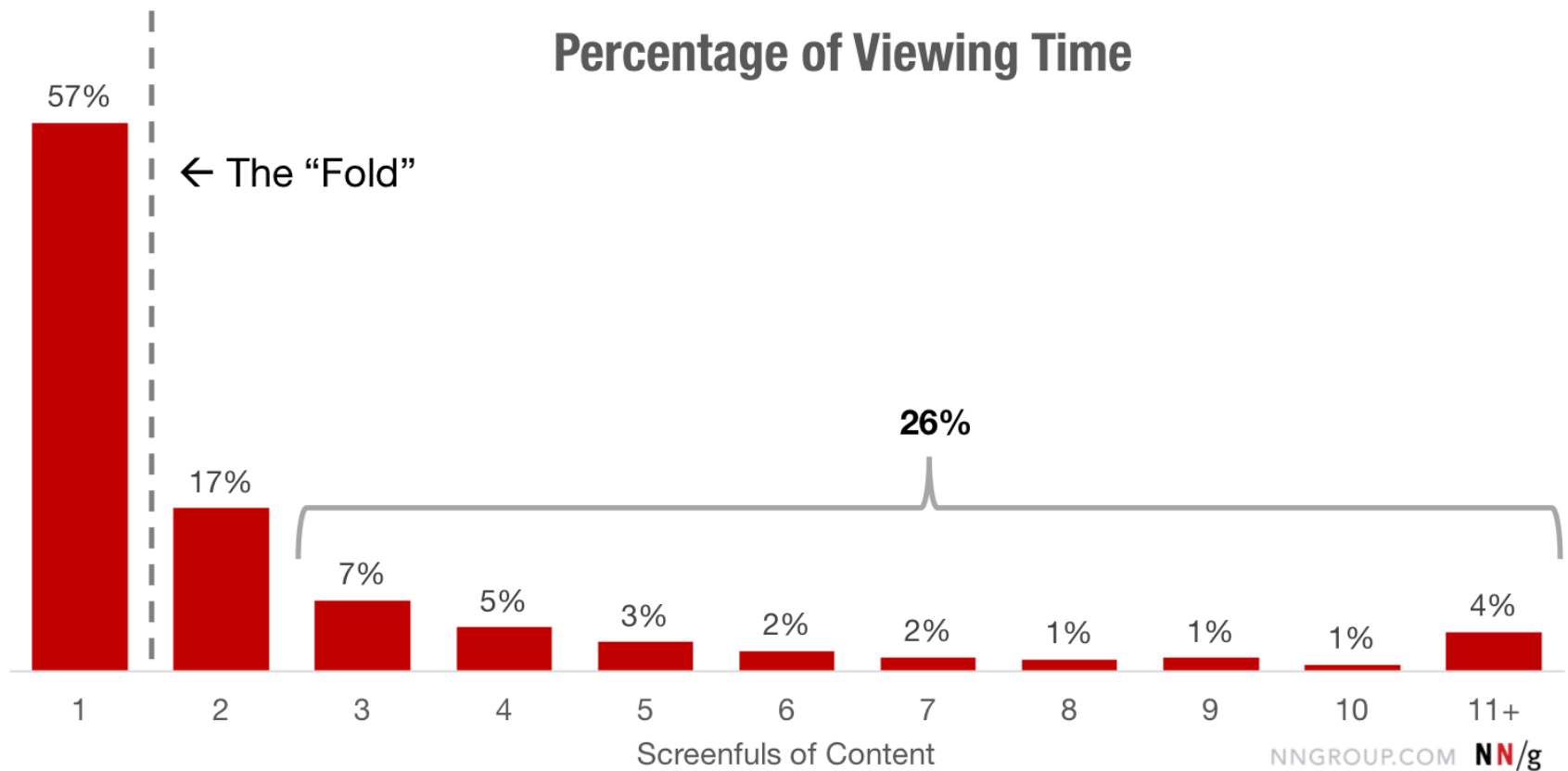
- **Typography:** The Golden Ratio can also be applied to typography by determining font sizes and line heights.
- **Spacing and Margins:** When determining spacing between elements, such as margins and padding, the Golden Ratio can guide the dimensions.
- **Visual Elements:** The Golden Ratio can be used to size images, buttons, and other visual elements. By ensuring that these elements adhere to the ratio, designers can create a cohesive and aesthetically pleasing design.

The golden ratio in web design

- **Focal Points:** The Golden Ratio can help in positioning focal points within a design. By placing key elements (like call-to-action buttons or important images) at points that align with the Golden Ratio, designers can draw attention to them effectively.
- Use it to achieve:
 - Aesthetic Appeal (including better UX).
 - Guided Composition (a clear guideline for designers).
 - Enhanced Readability (readability and a clear visual hierarchy).

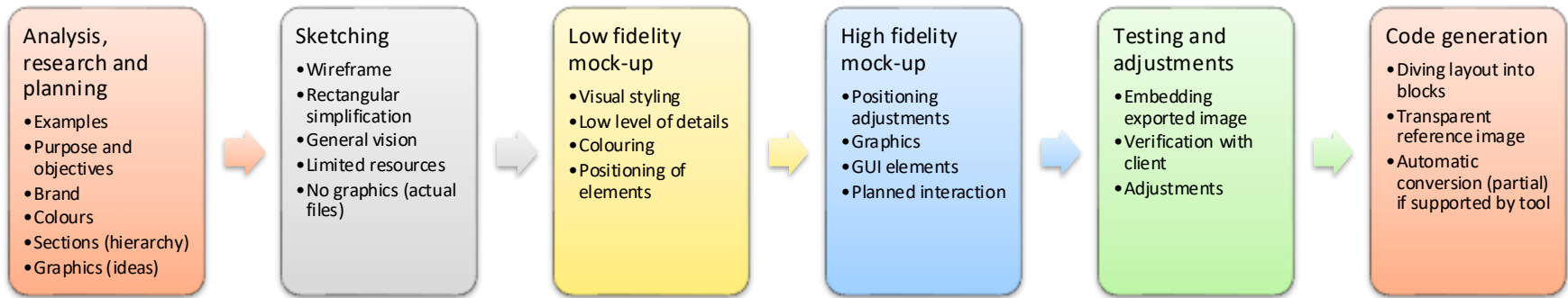
Website scrolling – The average fold

Nielsen Norman Group Nielsen Norman Group nngroup.com



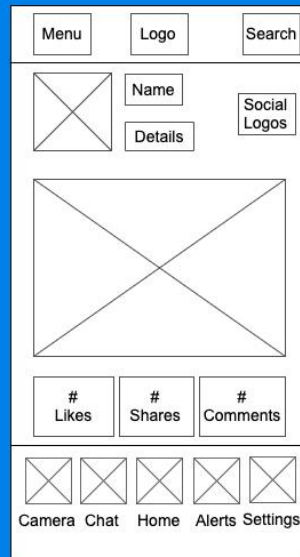
WEB LAYOUT CREATION

Transformation

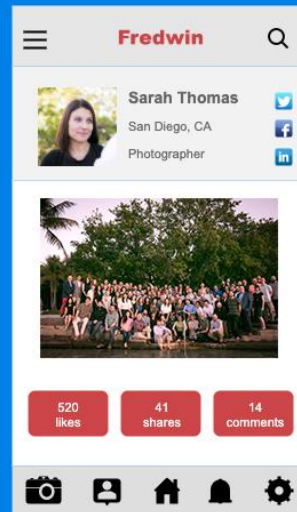


Artifacts delivered during web-development

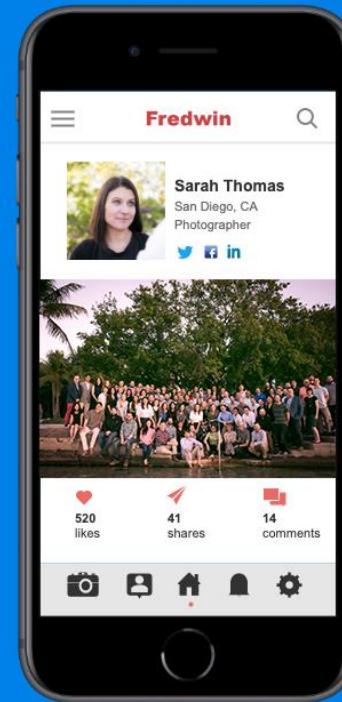
See: <https://www.aha.io>



Wireframe



Mockup

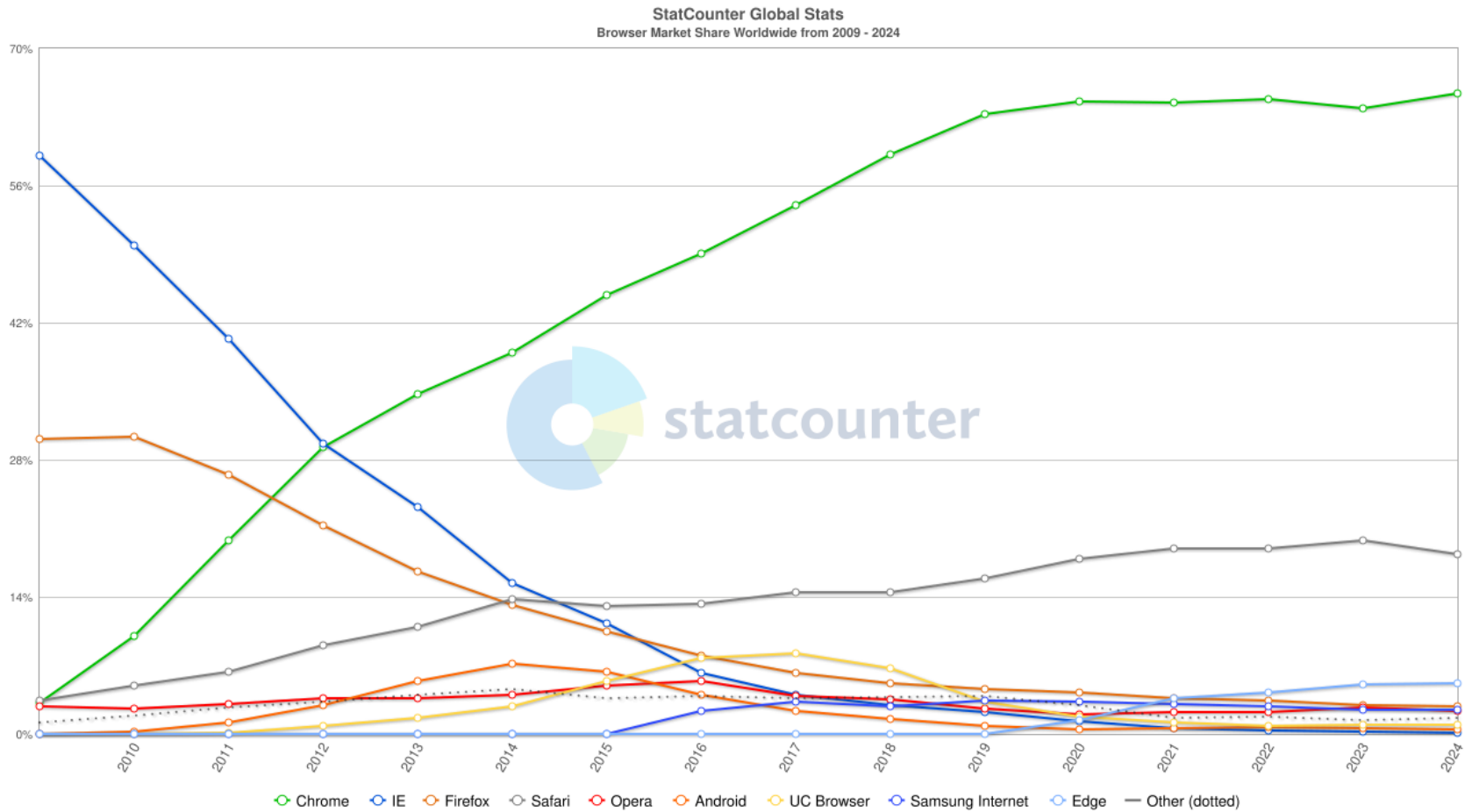


Prototype

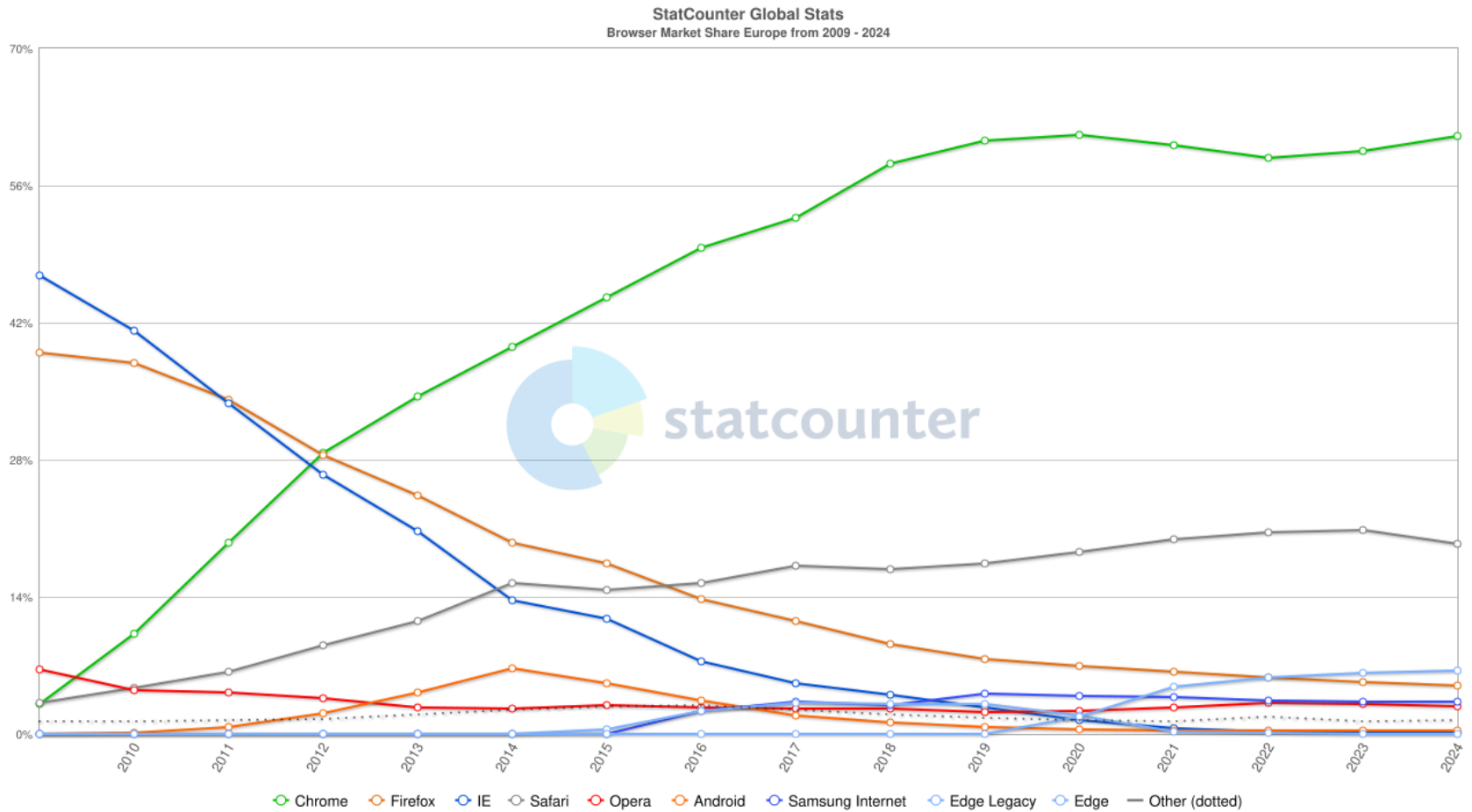
Hints

- Reuse components e.g. <https://design-system.w3.org>
- Look for layout examples e.g.
https://www.w3schools.com/css/css_templates.asp
<https://nicepage.com/css-templates>
- Try to use supporting tools e.g.
<https://www.lambdatest.com/blog/best-css-grid-layout-generators/>
- For positioning use grid (multiple columns) and snap elements to grid (limited by granulation, but it's a natural phenomenon)
- Go from top to bottom (according to the hierarchy of elements, not their vertical position)
- Select supported browsers (and used during testing)

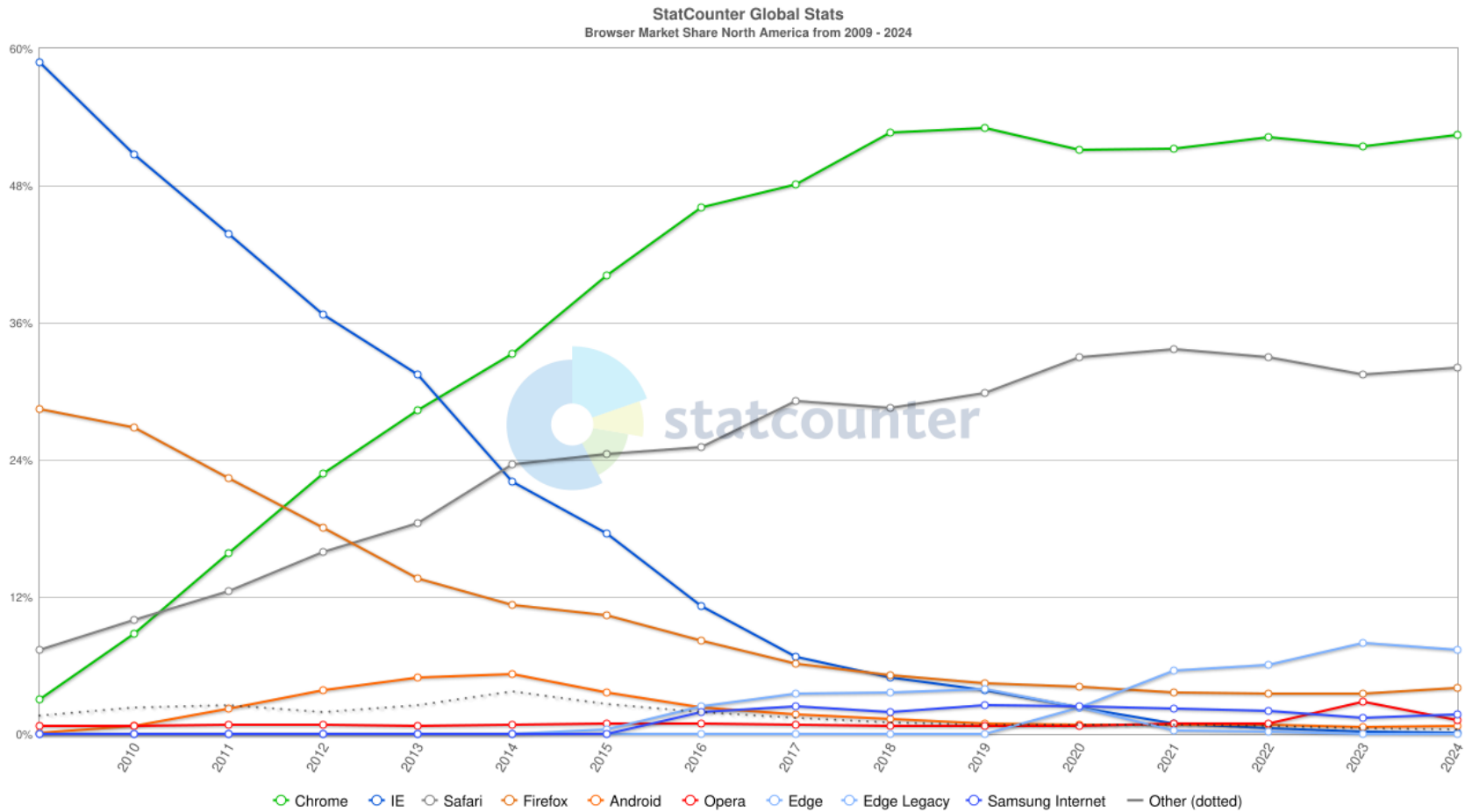
Browsers stats - Worldwide



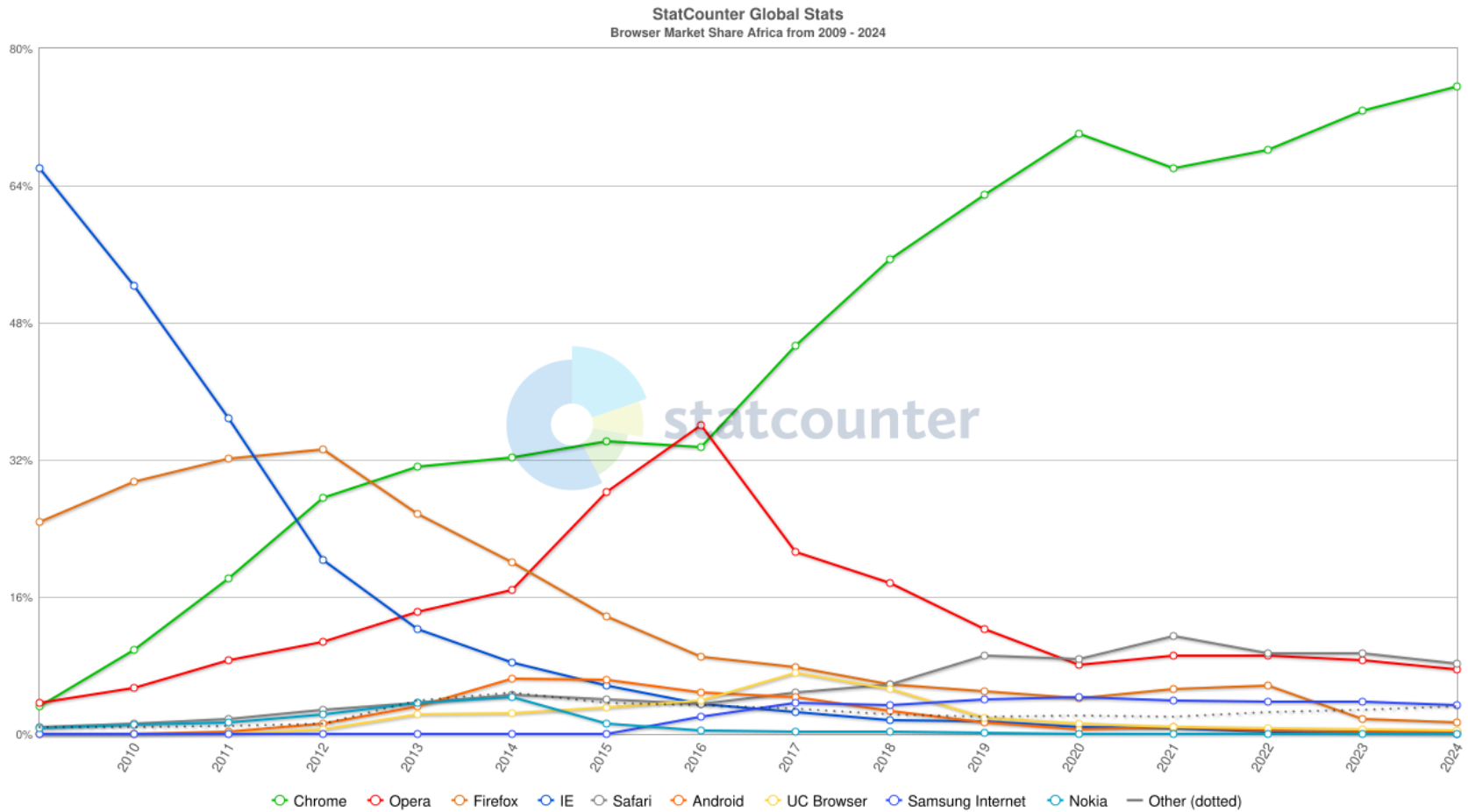
Browsers stats - EU



Browsers stats – North America



Browsers stats – Africa



Green-field layout vs Integration

Green-field

- No structure enforced by the other elements/systems
- Simpler structure
- Flexible development
- More choices to make
- No enforced principles
- Potential issues with further enhancements

Integration

- Structure enforced by the other elements/systems e.g. CMS/F
- More complex structure
- Development redefined by the limited/expectations
- Less choices to make
- Possible enforcement of web design principles
- Easier integration with other systems

CSS LAYOUTS

Basic types of CSS layouts

- There are 5 basic CSS layout types:
 - Natural flow
 - Positioned
 - Floats
 - Flexbox
 - Grid

Basic types of CSS layouts

- Each layout can be effectively used in selected scenarios, being ineffective in the other ones
- They also differ in simplicity, popularity (a rule of thumb – go with the crowd) or tools supporting them
- Let's examine examples at MDN:
https://developer.mozilla.org/en-US/docs/Learn/CSS/CSS_layout

Normal flow – Idea

- A default way by which elements are laid out the browser
- Based on the box model i.e. padding, border and margin is added to the content
- Block elements fill available inline parent's space, vertical dimension grows to accommodate the content
- The size of inline elements equals to the size of their content

Normal flow (MDN) – HTML

```
<h1>Basic document flow</h1>
<p>
  I am a basic block level element. My adjacent block level elements sit on new
  lines below me.
</p>
<p>
  By default we span 100% of the width of our parent element, and we are as tall
  as our child content. Our total width and height is our content + padding +
  border width/height.
</p>
<p>
  We are separated by our margins. Because of margin collapsing, we are
  separated by the width of one of our margins, not both.
</p>
<p>
  Inline elements <span>like this one</span> and <span>this one</span> sit on
  the same line along with adjacent text nodes, if there is space on the same
  line. Overflowing inline elements will
  <span>wrap onto a new line if possible (like this one containing text)</span>,
  or just go on to a new line if not, much like this image will do:
  
</p>
```

Normal flow (MDN) - CSS

```
body {  
  width: 500px;  
  margin: 0 auto;  
}
```

```
p {  
  background: rgba(255, 84, 104, 0.3);  
  border: 2px solid rgb(255, 84, 104);  
  padding: 10px;  
  margin: 10px;  
}
```

```
span {  
  background: white;  
  border: 1px solid black;  
}
```

Normal flow (MDN) – Layout

Basic document flow

I am a basic block level element. My adjacent block level elements sit on new lines below me.

By default we span 100% of the width of our parent element, and we are as tall as our child content. Our total width and height is our content + padding + border width/height.

We are separated by our margins. Because of margin collapsing, we are separated by the width of one of our margins, not both.

Inline elements like this one and this one sit on the same line along with adjacent text nodes, if there is space on the same line. Overflowing inline elements will wrap onto a new line if possible (like this one containing text), or just go on to a new line if not, much like this image will do:



Positioned – Idea

- CSS offers five main positioning types:
 - static,
 - relative,
 - absolute,
 - fixed,
 - and sticky.
- Each type affects how an element is positioned in relation to its normal flow, its parent container, or the viewport.

Positioned – Idea

- **Static:** The default positioning; elements are positioned according to the normal document flow.

```
<p class="positioned">...</p>
```

```
.positioned {  
  position: static;  
  background: yellow;  
}
```

Static positioning

I am a basic block level element. My adjacent block level elements sit on new lines below me.

By default we span 100% of the width of our parent element, and our are as tall as our child content. Our total width and height is our content + padding + border width/height.

We are separated by our margins. Because of margin collapsing, we are separated by the width of one of our margins, not both.

inline elements **like this one** and **this one** sit on the same line as one another, and adjacent text nodes, if there is space on the same line. Overflowing inline elements **wrap onto a new line if possible — like this one containing text**, or just go on to a new line if not, much like this image will do:



Positioned – Idea

- **Relative:** Positioned relative to its original position; can be adjusted using top, right, bottom, and left properties.

```
<p class="positioned">...</p>
```

Relative positioning

```
.positioned {  
  position: relative;  
  background: yellow;  
  top: 30px;  
  left: 30px;  
}
```

I am a basic block level element. My adjacent block level elements sit on new lines below me.

By default we span 100% of the width of our parent element, and we are as tall as our child content. Our total width and height is our content + padding + border width/height.

We are separated by our margins. Because of margin collapsing, we are separated by the width of one of our margins, not both.

Inline elements like this one and this one sit on the same line as one another, and adjacent text nodes, if there is space on the same line. Overflowing inline elements wrap onto a new line if possible — like this one containing text, or just go on to a new line if not, much like this image will do:



Positioned – Idea

- **Absolute:** Removed from the normal flow and positioned relative to the nearest positioned ancestor (not static); if none exists, it positions relative to the initial containing block (viewport).

```
<p class="positioned">...</p>
```

```
.positioned {  
  position: absolute;  
  background: yellow;  
  top: 30px;  
  left: 30px;  
}
```

Absolute positioning

By default we span 100% of the width of our parent element, and we are as tall as our child content. Our total width and height is our content + padding + border width/height.

I am a basic block level element. My adjacent block level elements sit on new lines below me.

We are separated by our margins. Because of margin collapsing, we are separated by the width of one of our margins, not both.

inline elements **like this one** and **this one** sit on the same line as one another, and adjacent text nodes, if there is space on the same line. Overflowing inline elements **wrap onto a new line if possible -- like this one containing text**, or just go on to a new line if not, much like this image will do:



Positioned – Idea

- **Fixed:** Removed from the document flow and positioned relative to the viewport; remains in the same position even when the page is scrolled.

```
<h1>Fixed positioning</h1>
h1 {
  position: fixed;
  top: 0;
  width: 500px;
  margin-top: 0;
  background: white;
  padding: 10px;
}
```

Fixed positioning

On new lines below me.

I'm not positioned any more.

We are separated by our margins. Because of margin collapsing, we are separated by the width of one of our margins, not both.

Inline elements **like this one** and **this one** sit on the same line as one another, and adjacent text nodes, if there is space on the same line. Overflowing inline elements **wrap onto a new line if possible — like this one containing text**, or just go on to a new line if not, much like this image will do:



Positioned – Idea

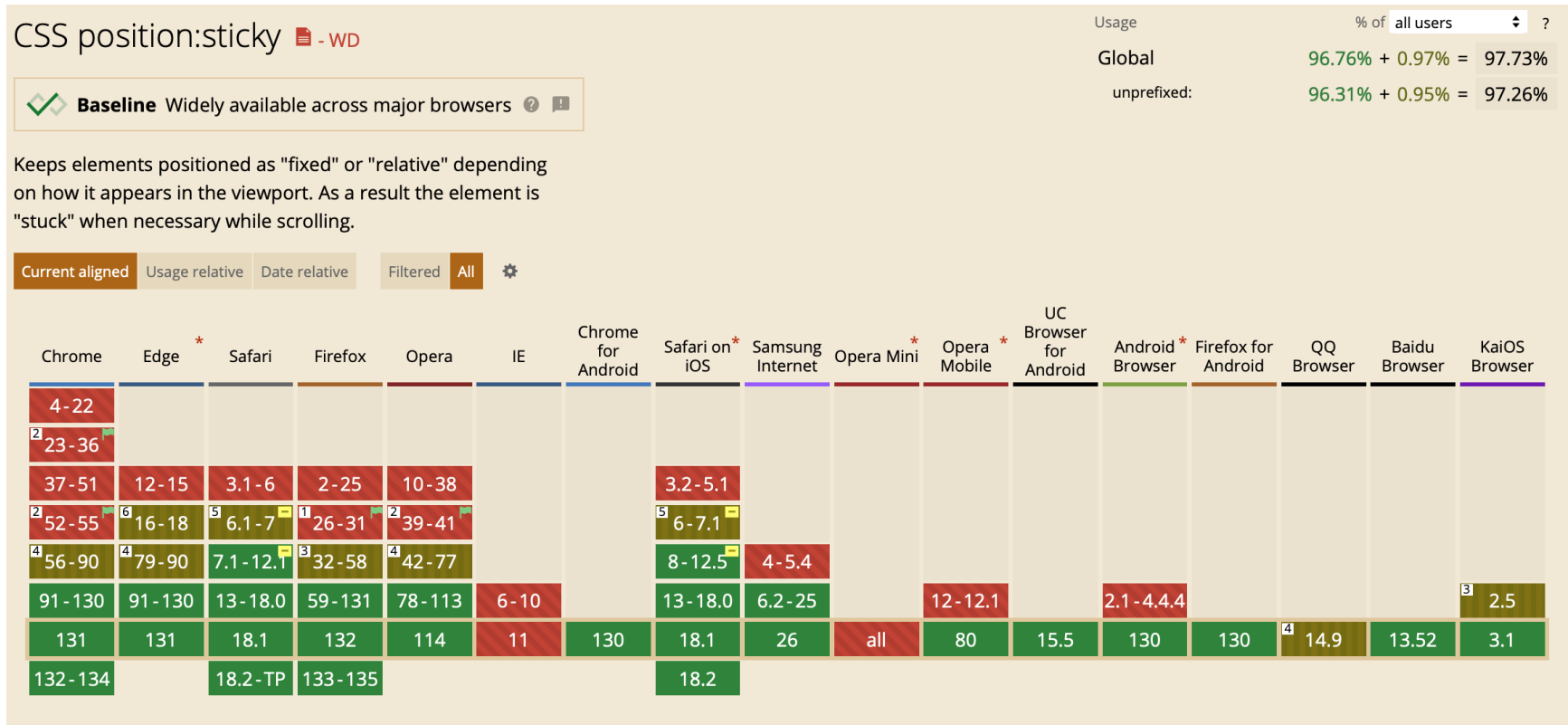
- **Sticky:** A hybrid of relative and fixed positioning; it behaves like relative until a specified scroll position is reached, then it becomes fixed.

```
<div class="positioned">Sticky</div>
.positioned {
  background: rgb(255 84 104 / 30%);
  border: 2px solid rgb(255 84 104);
  padding: 10px;
  margin: 10px;
  border-radius: 5px;
  position: sticky;
  top: 30px;
  left: 30px;
}
```

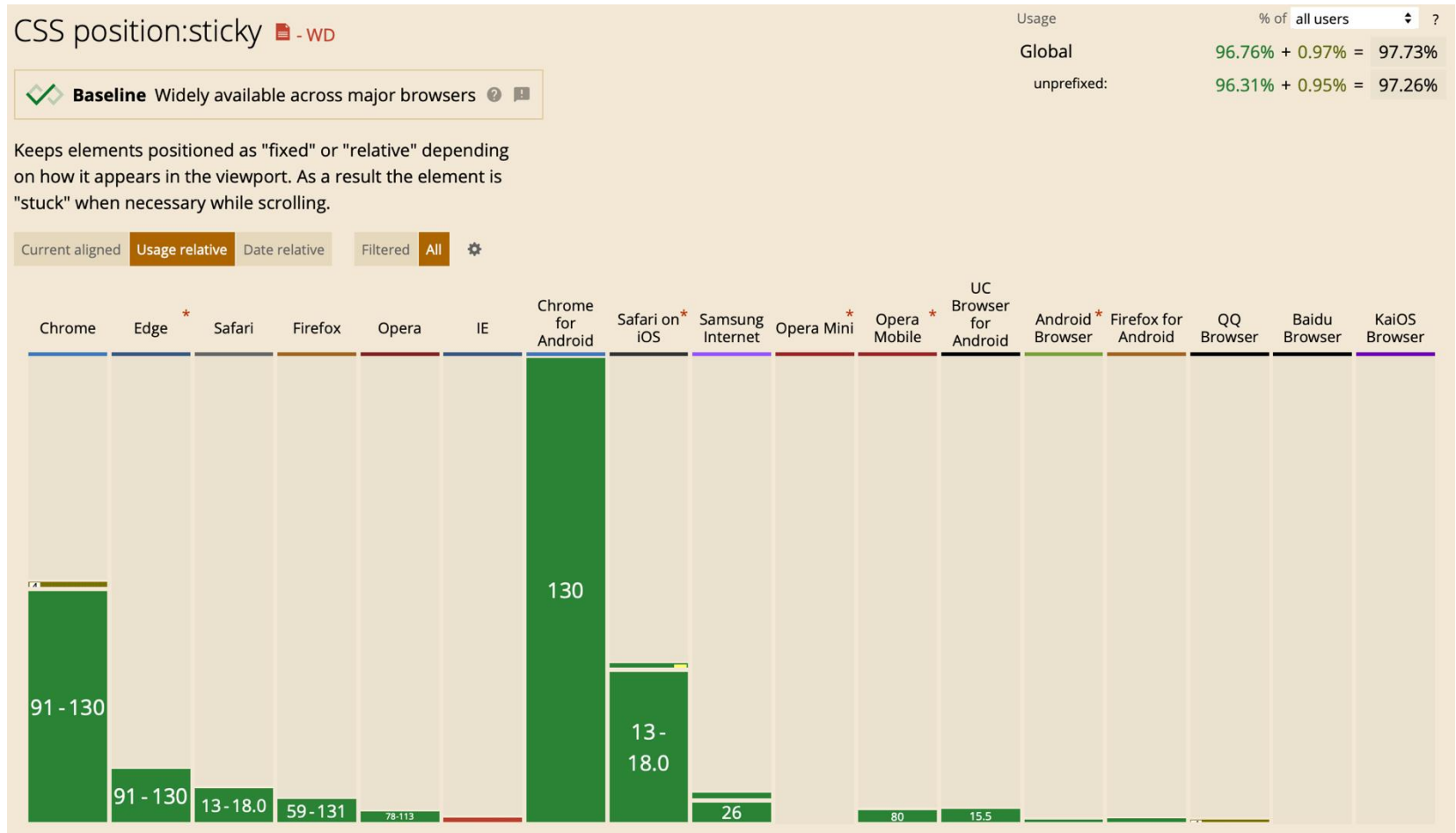
Nam vulputate diam nec tempor bibendum. Donec luctus augue eget malesuada ultrices. Phasellus turpis est, posuere sit amet dapibus ut, facilisis sed Nam id risus quis ante semper consectetur eget aliquam lorem. Vivamus tristique elit dolor, sed pretium metus suscipit vel. Mauris ultricies lectus sed lobortis finibus. Vivamus eu urna eget velit cursus viverra quis vestibulum sem. Aliquam tincidunt eget purus in interdum. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla luctus aliquam dolor, eu lacinia lorem placerat vulputate. Duis felis orci, pulvinar id metus ut, rutrum luctus orci. Cras porttitor imperdiet nunc, at ultricies tellus laoreet sit amet. Sed auctor cursus massa at porta. Integer ligula ipsum, tristique sit amet orci vel, viverra egestas ligula. Curabitur vehicula tellus neque, ac ornare ex malesuada et. In vitae convallis lacus. Aliquam erat volutpat. Suspendisse ac imperdiet turpis. Aenean finibus sollicitudin eros pharetra congue. Duis ornare egestas augue ut luctus. Proin blandit quam nec lacus varius commodo et a urna. Ut id ornare felis, eget fermentum sapien.

Positioned (sticky property) – Support



Positioned (sticky property) – Support



Floats – Idea

- Originally created for floating images
- With float property elements can float inside parent element, with text wrapping around
- For a while used to create multicolumn layouts
- Replaced by the flexbox and grid layouts, currently mainly used for the original purposes
- Still possible to be used, however is considered to be a legacy technique

Floats (MDN) – HTML

```
<h1>Simple float example</h1>
<div class="box">Float</div>
<p>
  Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla luctus aliquam
  dolor, eu lacinia lorem placerat vulputate. Duis felis orci, pulvinar id metus
  ut, rutrum luctus orci. Cras porttitor imperdiet nunc, at ultricies tellus
  laoreet sit amet.
</p>
<p>
  Sed auctor cursus massa at porta. Integer ligula ipsum, tristique sit amet
  orci vel, viverra egestas ligula. Curabitur vehicula tellus neque, ac ornare
  ex malesuada et. In vitae convallis lacus. Aliquam erat volutpat. Suspendisse
  ac imperdiet turpis. Aenean finibus sollicitudin eros pharetra congue. Duis
  ornare egestas augue ut luctus. Proin blandit quam nec lacus varius commodo et
  a urna. Ut id ornare felis, eget fermentum sapien.
</p>
<p>
  Nam vulputate diam nec tempor bibendum. Donec luctus augue eget malesuada
  ultrices. Phasellus turpis est, posuere sit amet dapibus ut, facilisis sed
  est. Nam id risus quis ante semper consectetur eget aliquam lorem. Vivamus
  tristique elit dolor, sed pretium metus suscipit vel. Mauris ultricies lectus
  sed lobortis finibus. Vivamus eu urna eget velit cursus viverra quis
  vestibulum sem. Aliquam tincidunt eget purus in interdum. Cum sociis natoque
  penatibus et magnis dis parturient montes, nascetur ridiculus mus.
</p>
```

Floats (MDN) - CSS

```
body {  
  width: 90%;  
  max-width: 900px;  
  margin: 0 auto;  
  font: 0.9em/1.2 Arial, Helvetica, sans-serif;  
}
```

```
.box {  
  float: left;  
  margin-right: 15px;  
  width: 150px;  
  height: 100px;  
  border-radius: 5px;  
  background-color: rgb(207, 232, 220);  
  padding: 1em;  
}
```

Floats (MDN) – Layout

Simple float example

Float

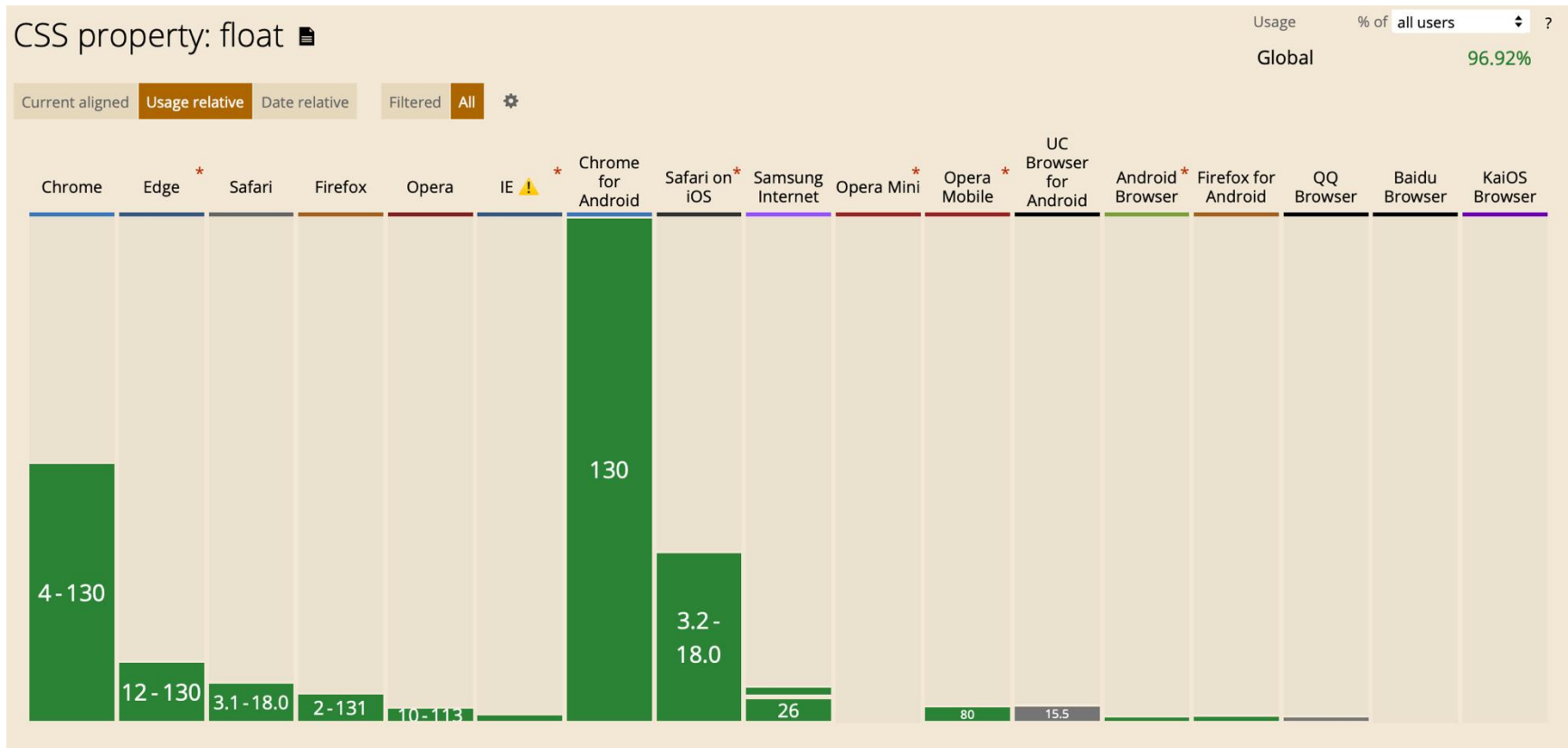
Lorem ipsum dolor sit amet, consectetur adipiscing elit. Nulla luctus aliquam dolor, eu lacinia lorem placerat vulputate. Duis felis orci, pulvinar id metus ut, rutrum luctus orci. Cras porttitor imperdiet nunc, at ultricies tellus laoreet sit amet.

Sed auctor cursus massa at porta. Integer ligula ipsum, tristique sit amet orci vel, viverra egestas ligula. Curabitur vehicula tellus neque, ac ornare ex malesuada et. In vitae convallis lacus. Aliquam erat volutpat. Suspendisse ac imperdiet turpis. Aenean finibus sollicitudin eros pharetra congue. Duis ornare egestas augue ut

luctus. Proin blandit quam nec lacus varius commodo et a urna. Ut id ornare felis, eget fermentum sapien.

Nam vulputate diam nec tempor bibendum. Donec luctus augue eget malesuada ultrices. Phasellus turpis est, posuere sit amet dapibus ut, facilisis sed est. Nam id risus quis ante semper consectetur eget aliquam lorem. Vivamus tristique elit dolor, sed pretium metus suscipit vel. Mauris ultricies lectus sed lobortis finibus. Vivamus eu urna eget velit cursus viverra quis vestibulum sem. Aliquam tincidunt eget purus in interdum. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

Floats – Support



Flexbox – Idea

- One-dimensional layout used to arrange items in rows or columns
- Items flex filling available space i.e. expanding or shrinking depending on the size of space
- The children element of a container could take an equal amount of available width/height

Flexbox (MDN) - Sample

Sample flexbox example

First article

Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Second article

Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Third article

Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Cray food truck brunch, XOXO +1 keffiyeh pickled chambray waistcoat ennui. Organic small batch paleo 8-bit. Intelligentsia umami wayfarers pickled, asymmetrical kombucha letterpress kitsch leggings cold-pressed squid chartreuse put a bird on it. Listicle pickled man bun cornhole heirloom art party.

Flexbox (MDN) – HTML

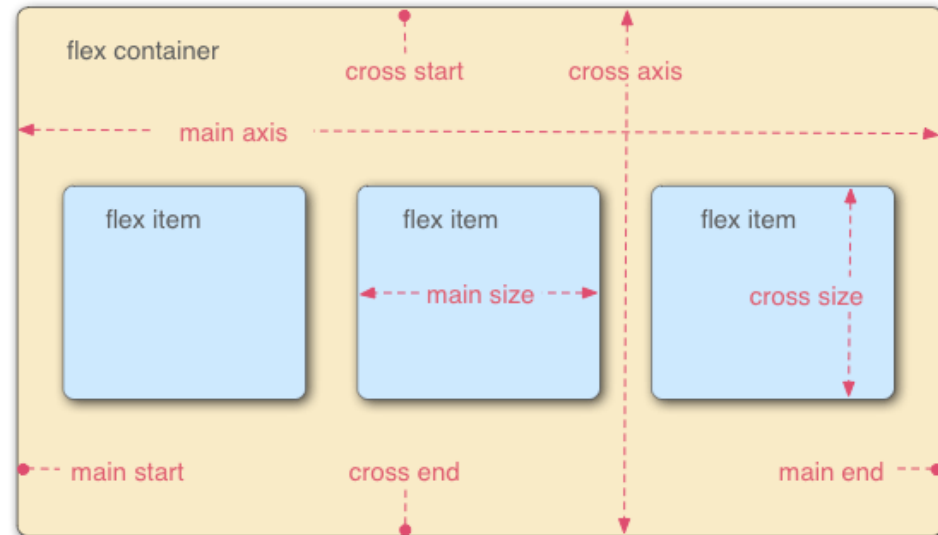
```
<section>
  <article>
    <h2>First article</h2>
    <p>Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack
portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled
sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund
twee, leggings salvia iPhone photo booth health goth gastropub hammock.</p>
  </article>
  <article>
    <h2>Second article</h2>
    <p>Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack
portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled
sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund
twee, leggings salvia iPhone photo booth health goth gastropub hammock.</p>
  </article>
  <article>
    <h2>Third article</h2>
    <p>Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack
portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled
sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund
twee, leggings salvia iPhone photo booth health goth gastropub hammock.</p>
    <p>Cray food truck brunch, XOXO +1 keffiyeh pickled chambray waistcoat ennui. Organic
small batch paleo 8-bit. Intelligentsia umami wayfarers pickled, asymmetrical kombucha
letterpress kitsch leggings cold-pressed squid chartreuse put a bird on it. Listicle pickled man
bun cornhole heirloom art party.</p>
  </article>
</section>
```


Flexbox (MDN) - CSS

```
section {  
  display: flex;  
}
```

Columns vs rows:
`flex-direction: column/row;`

Text wrapping:
`flex-wrap: wrap; (to Section)`
`flex: 200px; (to Article)`



Flexbox (MDN) – Layout

Sample flexbox example

First article

Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Second article

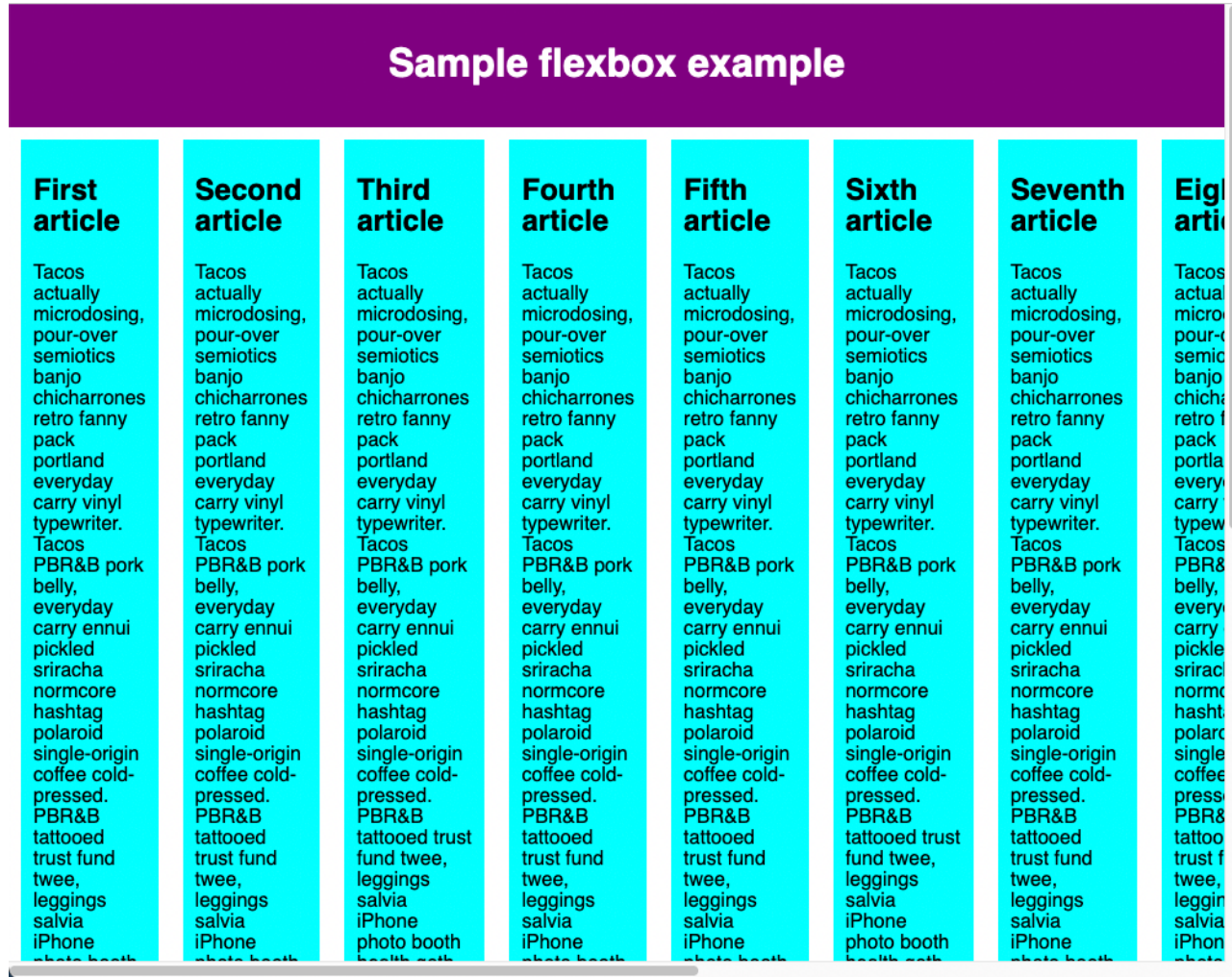
Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Third article

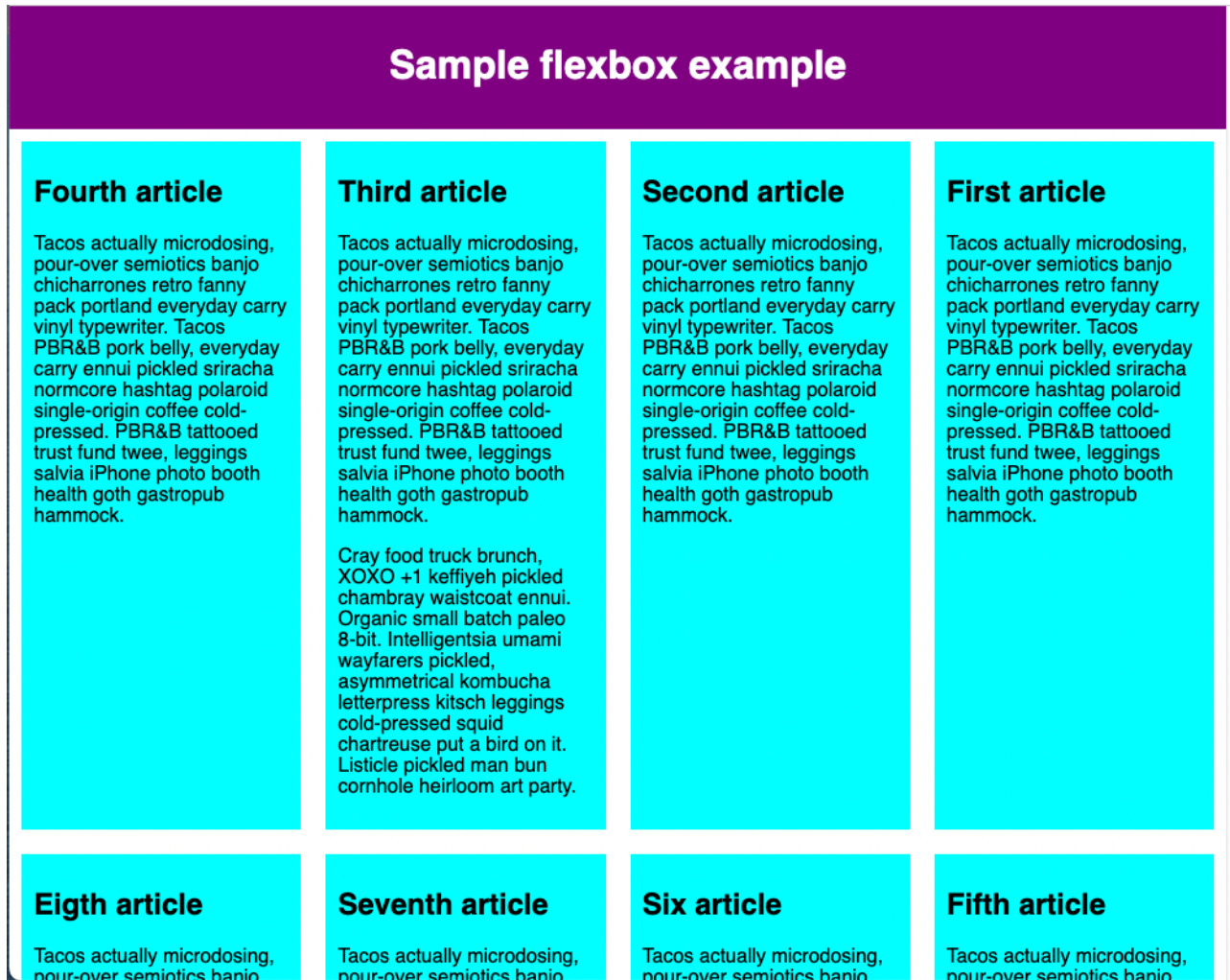
Tacos actually microdosing, pour-over semiotics banjo chicharrones retro fanny pack portland everyday carry vinyl typewriter. Tacos PBR&B pork belly, everyday carry ennui pickled sriracha normcore hashtag polaroid single-origin coffee cold-pressed. PBR&B tattooed trust fund twee, leggings salvia iPhone photo booth health goth gastropub hammock.

Cray food truck brunch, XOXO +1 keffiyeh pickled chambray waistcoat ennui. Organic small batch paleo 8-bit. Intelligentsia umami wayfarers pickled, asymmetrical kombucha letterpress kitsch leggings cold-pressed squid chartreuse put a bird on it. Listicle pickled man bun cornhole heirloom art party.

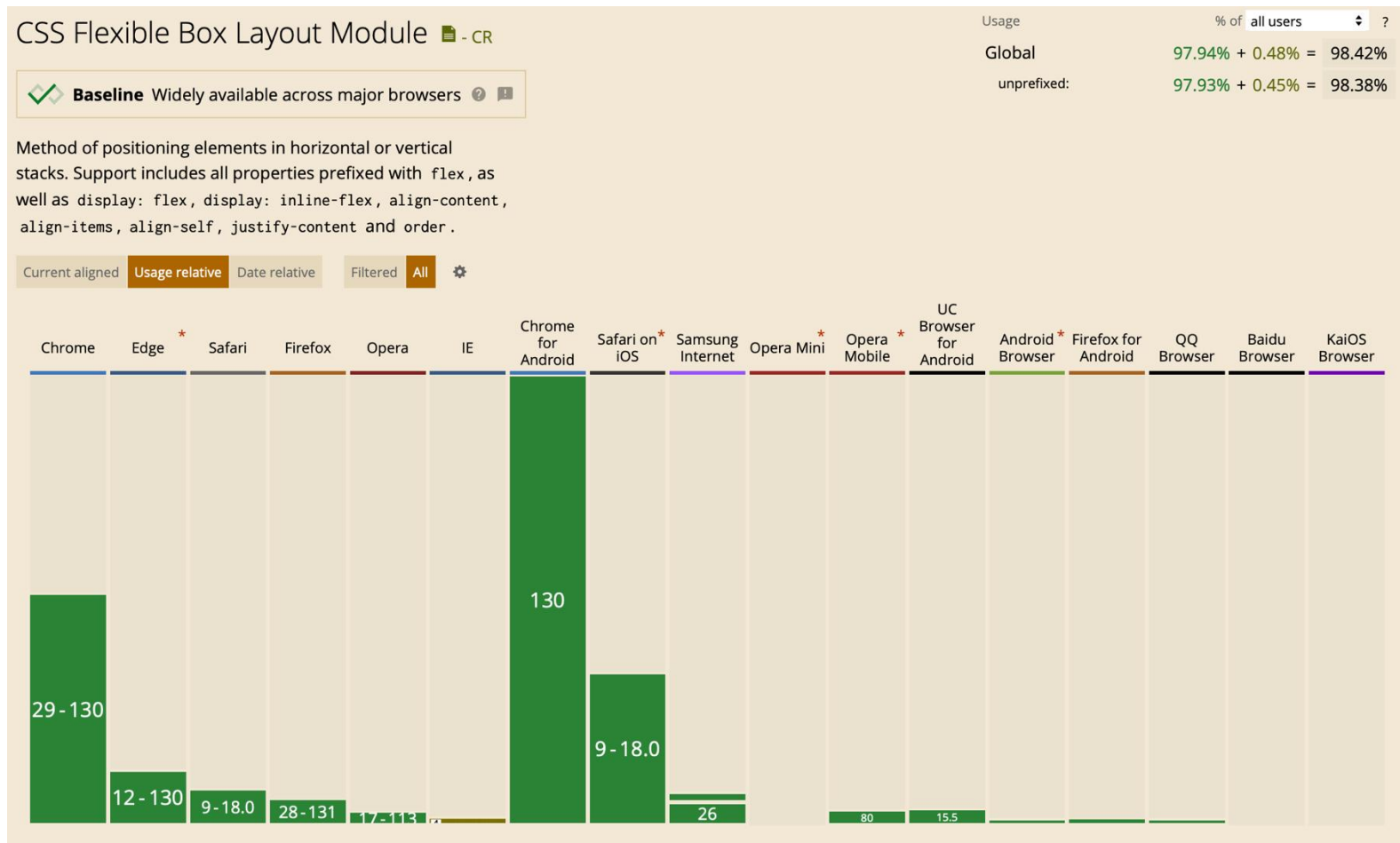
Flexbox (MDN) – Layout



Flexbox (MDN) – Layout



Flexbox – Support



Grid – Idea

- Two-dimensional layout (rows and columns)
- Commonly used for complex layouts
- The majority of layouts can be divided onto elements of grid
- Elements can span onto multiple "cells" i.e. grid doesn't have to be in a matrix form
- Grid can be combined with flexbox
- A proper replacement of table-based layouts

Grid (MDN) – HTML

```
<body>
  <h1>Simple grid example</h1>
  <div class="container">
    <div>One</div>
    <div>Two</div>
    <div>Three</div>
    <div>Four</div>
    <div>Five</div>
    <div>Six</div>
    <div>Seven</div>
  </div>
```

```
</body>
```

```
----
body {
  width: 90%;
  max-width: 900px;
  margin: 2em auto;
  font: .9em/1.2 Arial, Helvetica, sans-serif;
}
```

```
.container > div {
  border-radius: 5px;
  padding: 10px;
  background-color: rgb(207,232,220);
  border: 2px solid rgb(79,185,227);
}
```

Simple grid example

One
Two
Three
Four
Five
Six
Seven

Grid (MDN) - CSS

```
.container {  
  display: grid;  
  grid-template-columns: 200px 200px 200px;  
}  
---  
header {  
  grid-column: 1 / 3;  
  grid-row: 1;  
}  
article {  
  grid-column: 2;  
  grid-row: 2;  
}  
aside {  
  grid-column: 1;  
  grid-row: 2;  
}  
footer {  
  grid-column: 1 / 3;  
  grid-row: 3;  
}
```

One	Two	Three
Four	Five	Six
Seven		

This is my lovely blog

Other things

Nam vulputate diam nec tempor bibendum. Donec luctus augue eget malesuada ultrices. Phasellus turpis est, posuere sit amet dapibus ut, facilisis sed est.

My article

Duis felis orci, pulvinar id metus ut, rutrum luctus orci. Cras porttitor imperdiet nunc, at ultricies tellus laoreet sit amet. Sed auctor cursus massa at porta. Integer ligula ipsum, tristique sit amet orci vel, viverra egestas ligula. Curabitur vehicula tellus neque, ac ornare ex malesuada et. In vitae convallis lacus. Aliquam erat volutpat. Suspendisse ac imperdiet turpis. Aenean finibus sollicitudin eros pharetra congue. Duis ornare egestas augue ut luctus. Proin blandit quam nec lacus varius commodo et a urna. Ut id ornare felis, eget fermentum sapien.

Nam vulputate diam nec tempor bibendum. Donec luctus augue eget malesuada ultrices. Phasellus turpis est, posuere sit amet dapibus ut, facilisis sed est. Nam id risus quis ante semper consectetur eget aliquam lorem. Vivamus tristique elit dolor, sed pretium metus suscipit vel. Mauris ultricies lectus sed lobortis finibus. Vivamus eu urna eget velit cursus viverra quis vestibulum sem. Aliquam tincidunt eget purus in interdum. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus.

Contact me@example.com

Grid – Support



Which layout should I choose?

- Prefer newer layouts i.e. flexbox or grid, and avoid legacy ones
- Consider if:
 - elements are distributed in one dimensions (flexbox) or in two (grid)
 - As a basis should be taken content (flexbox), or layout (grid)
- It is possible to combine both (by embedding, not mixing)

It's not only about the visuals #1

- **Consistent Branding:** Use consistent colours, fonts, and imagery that align with your brand identity. This helps create a cohesive look and feel across your website.
- **Hierarchy and Layout:** Establish a clear visual hierarchy using size, colour, and spacing. Important elements should stand out, guiding users through the content naturally.

It's not only about the visuals #2

- **Navigation:** Design intuitive navigation that is easy to use. Keep menus simple and organised and ensure that users can find what they need quickly.
- **Call to Action (CTA):** Make your CTAs prominent and compelling. Use contrasting colours and clear language to encourage users to take action.

It's not only about the visuals #3

- **Load Speed:** Optimize images and minimize code to improve page load speed. A fast-loading website enhances user experience and can positively impact SEO.
- **SEO Best Practices:** Incorporate SEO techniques, such as using descriptive URLs, optimizing meta tags, and including relevant keywords, to improve your website's visibility in search engines.

It's not only about the visuals #4

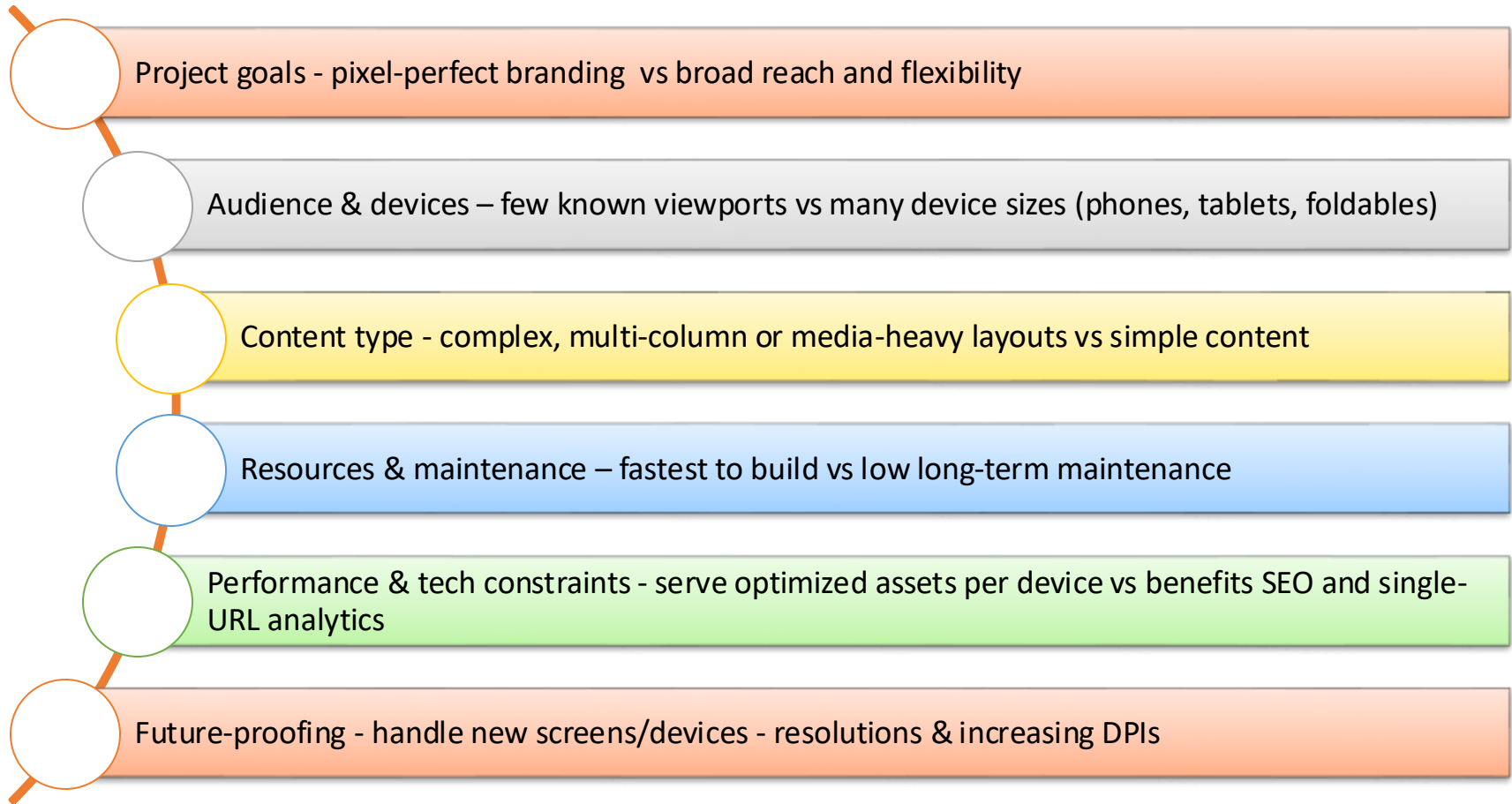
- **Accessibility:** Ensure your website is accessible to all users, including those with disabilities. Use alt text for images, proper heading structures, and ensure keyboard navigation is possible.
- **Visual Elements:** Use high-quality images and graphics that enhance your content. Avoid using too many stock images; instead, opt for original visuals when possible.

It's not only about the visuals #5

- **Testing and Feedback:** Regularly test your website on different devices and browsers. Gather user feedback to identify areas for improvement and make necessary adjustments.
- **Analytics:** Use web analytics tools to track user behaviour and gather insights. This data can help you make informed decisions about design and content improvements.

DESIGNS

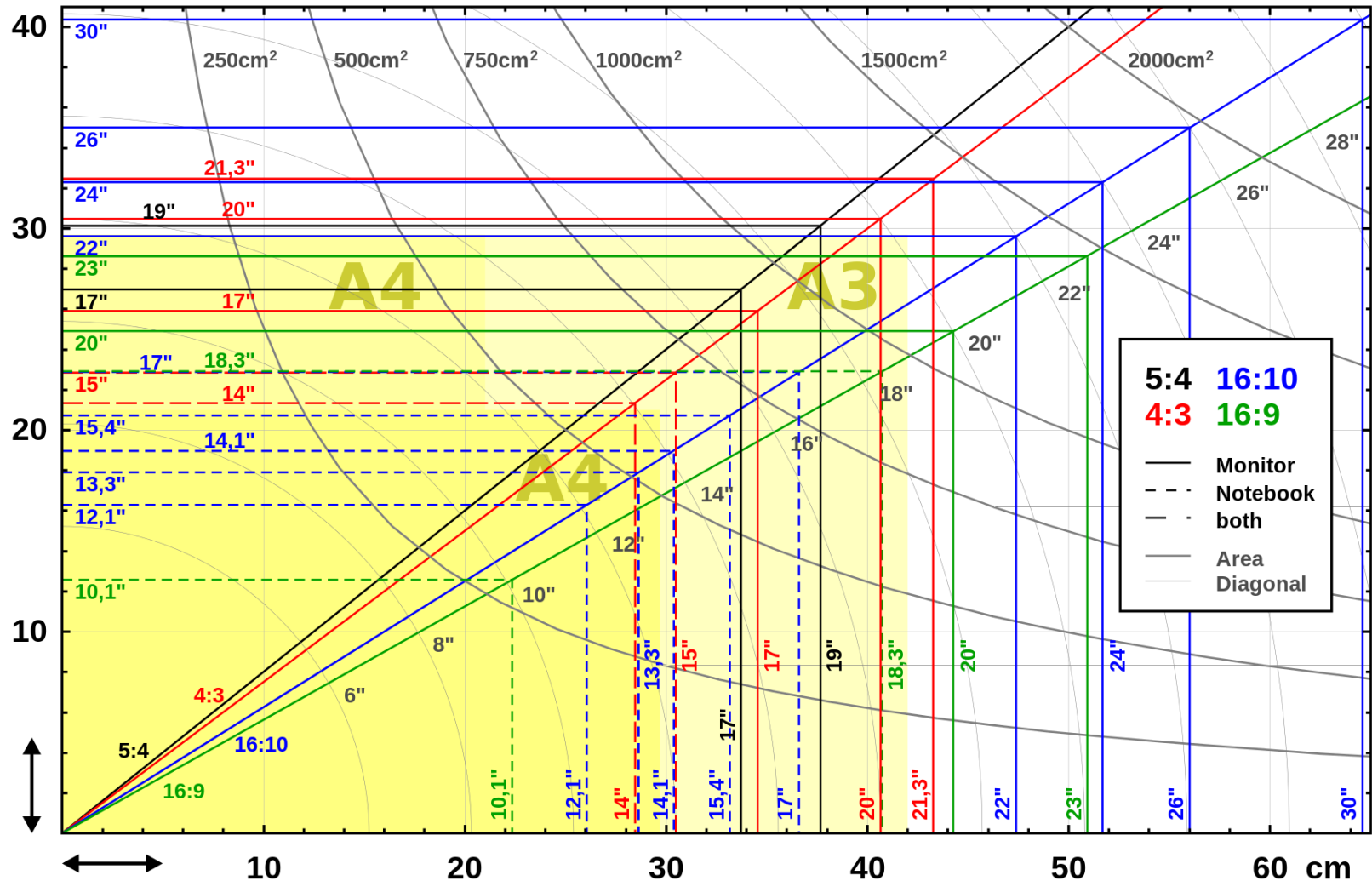
Web Designs – Factors



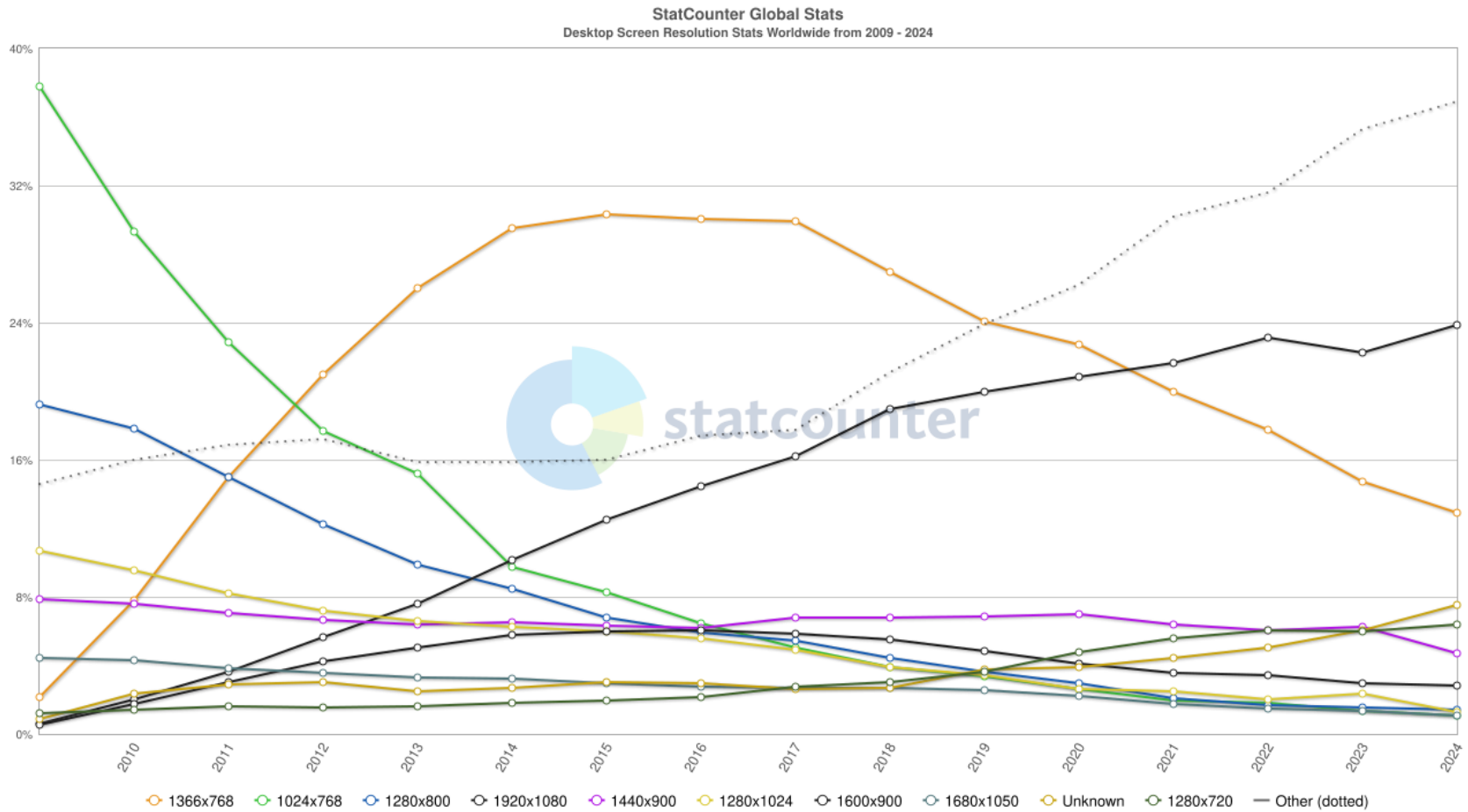
Screen size – iPhone (source idev101.com)



Screen size – displays



Desktop resolution – Worldwide Stats



Fixed (Static) Design

- Uses fixed-width containers (usually pixels). Content and elements stay the same size regardless of screen size.
- Use for: Simple brochure sites, designs optimized for a single target width, or when precise control over layout is required.
- Pros: Predictable layout, easier to design pixel-perfect pages, less CSS complexity.
- Cons: Poor on small/large screens (horizontal scrolling or large margins), not future-proof for varied devices.

HTML Example

```
<div class="card fixed">
  
  <div class="info">
    <h2>Luke Skywalker</h2>
    <p>Farm kid turned galaxy-saving intern who
learned sword-fighting, piloting, and emotional
baggage management in one semester.</p>
    <button>Contact</button>
  </div>
</div>
```

CSS

```
.card.fixed {  
  width: 800px;  
  border: 1px solid #ccc;  
  padding: 16px;  
  overflow: auto;  
}  
.card.fixed img {  
  float: left;  
  width: 150px;  
  height: 150px;  
  object-fit: cover;  
  margin-right: 16px;  
}  
.card.fixed .info { width: 600px; }  
.card.fixed button {  
  width: 120px;  
  height: 40px;  
}
```

Fluid (Liquid) Design

- Uses proportional units (percentages) so content stretches and shrinks continuously to fill the viewport—no fixed-width containers.
- Use for: When content should maximize available space across arbitrary viewport sizes, or as part of responsive approaches.
- Pros: Smooth scaling across sizes, simple implementation for flexible spacing.
- Cons: Can produce awkward line lengths or element sizes at extremes; often combined with responsive breakpoints to avoid issues.

HTML Example

```
<div class="card fluid">
  
  <div class="info">
    <h2>Princess Leia</h2>
    <p>Diplomatic mastermind who can command a fleet,
lead a rebellion, and deliver the perfect sarcastic
burn—often all before breakfast.</p>
    <button>Contact</button>
  </div>
</div>
```

CSS

```
.card.fluid {  
  width: 90%;  
  max-width: 1200px;  
  border: 1px solid #ccc;  
  padding: 2%;  
  box-sizing: border-box;  
}  
.card.fluid img {  
  float: left;  
  width: 20%;  
  height: auto;  
  object-fit: cover;  
  margin-right: 2%;  
}  
.card.fluid .info { width: 75%; }  
.card.fluid button { width: 15%; min-width: 80px; }
```

Adaptive Design

- Multiple distinct layouts are created for specific viewport ranges. The site detects the user's screen size (server- or client-side) and serves the matching layout.
- Use for: Projects needing tailored experiences for certain device classes (e.g., separate mobile, tablet or desktop designs).
- Pros: Optimized, device-specific UX; performance gains by serving only what's needed per breakpoint.
- Cons: More design/development work (multiple layouts), harder to maintain many variants, gaps between breakpoints can occur.

HTML Example

```
<div class="card adaptive">
  
  <div class="info">
    <h2>Darth Vader</h2>
    <p>Black-suited executive with a heavy breathing
problem who enforces family rules using power-chokes
and dramatic wardrobe choices.</p>
    <button>Contact</button>
  </div>
</div>
```

CSS

```
/* Mobile preset (≤599px) */
@media (max-width: 599px) {
    .card.adaptive { padding: 12px; border: 1px solid #ccc; }
    .card.adaptive img { display: block; width: 72px; height: 72px; border-radius: 6px; margin:
0 auto 8px; }
    .card.adaptive .info { text-align: center; }
    .card.adaptive button { width: 90%; margin: 8px auto 0; display: block; }
}

/* Tablet preset (600-1023px) */
@media (min-width: 600px) and (max-width: 1023px) {
    .card.adaptive { display: flex; gap: 12px; padding: 16px; }
    .card.adaptive img { width: 120px; height: 120px; }
    .card.adaptive .info { flex: 1; }
    .card.adaptive button { width: 160px; }
}

/* Desktop preset (≥1024px) */
@media (min-width: 1024px) {
    .card.adaptive { display: grid; grid-template-columns: 150px 1fr; gap: 20px; padding: 20px;
}
    .card.adaptive img { width: 150px; height: 150px; }
    .card.adaptive .info { align-self: center; }
    .card.adaptive button { width: 140px; }
}
```

Responsive Design

- A single flexible layout that adapts fluidly across screen sizes using relative units (% , em, rem), flexible grids, images, and CSS media queries to change styles at breakpoints.
- Use for: Most modern websites aiming to support a wide range of devices with one codebase.
- Pros: Consistent experience across devices, maintainable single codebase, widely supported best practice.
- Cons: Can require careful performance tuning (images/resources) and thoughtful breakpoints; may be less optimized than adaptive for particular devices.

HTML Example

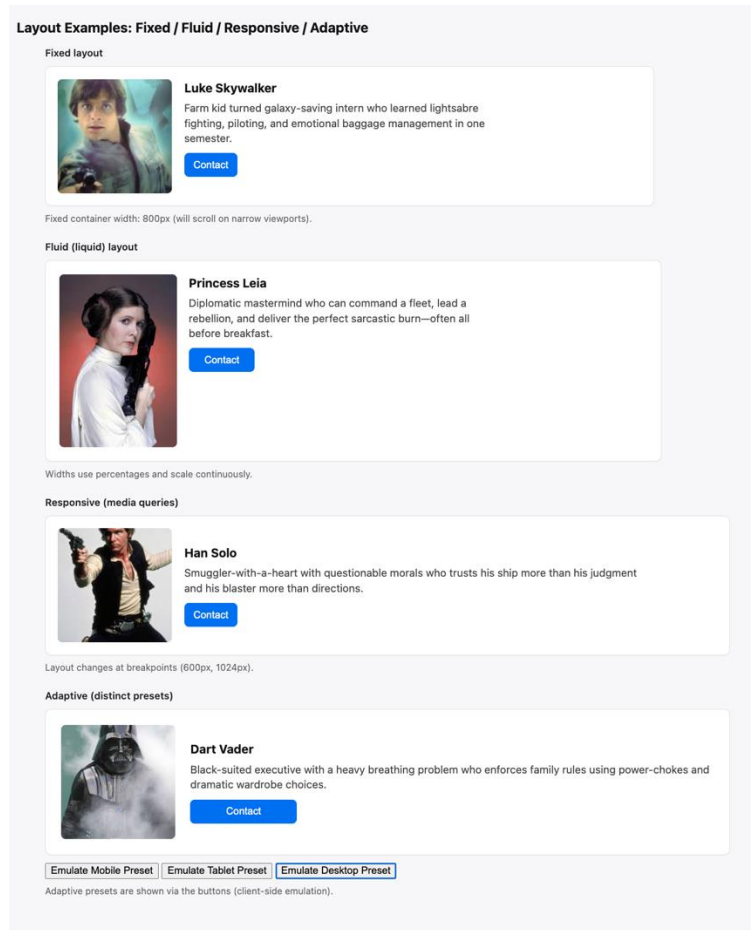
```
<div class="card responsive">
  
  <div class="info">
    <h2>Han Solo</h2>
    <p>Smuggler-with-a-heart-of-questionable-morals
who trusts his ship more than his judgment and his
blaster more than directions.</p>
    <button>Contact</button>
  </div>
</div>
```

Dynamic Documents and Front-end Web Development

```
.card.responsive {  
  border: 1px solid #ccc;  
  padding: 16px;  
  box-sizing: border-box;  
}  
/* Mobile default: stacked */  
.card.responsive img {  
  display: block;  
  width: 100%;  
  height: auto;  
  margin-bottom: 12px;  
}  
.card.responsive .info { width: 100%; }  
.card.responsive button { width: 100%; }  
  
/* Tablet / small desktop */  
@media (min-width: 600px) {  
  .card.responsive {  
    display: flex;  
    gap: 16px;  
    align-items: center;  
  }  
  .card.responsive img { width: 30%; max-width: 200px; margin: 0; }  
  .card.responsive .info { width: 70%; }  
  .card.responsive button { width: auto; }  
}  
  
/* Large screens */  
@media (min-width: 1024px) {  
  .card.responsive img { width: 150px; height: 150px; }  
}
```

CSS

The webpage



Experiments will be
done in
the laboratory

Focus on

In short, from a technical stance designs:

- Fixed – uses absolute pixel widths;
- Fluid – are based on percentages so elements scale continuously;
- Responsive – deploy flexible units plus media-query breakpoints to change layout;
- Adaptive – provide distinct, tuned presets per range.

Practical guidance

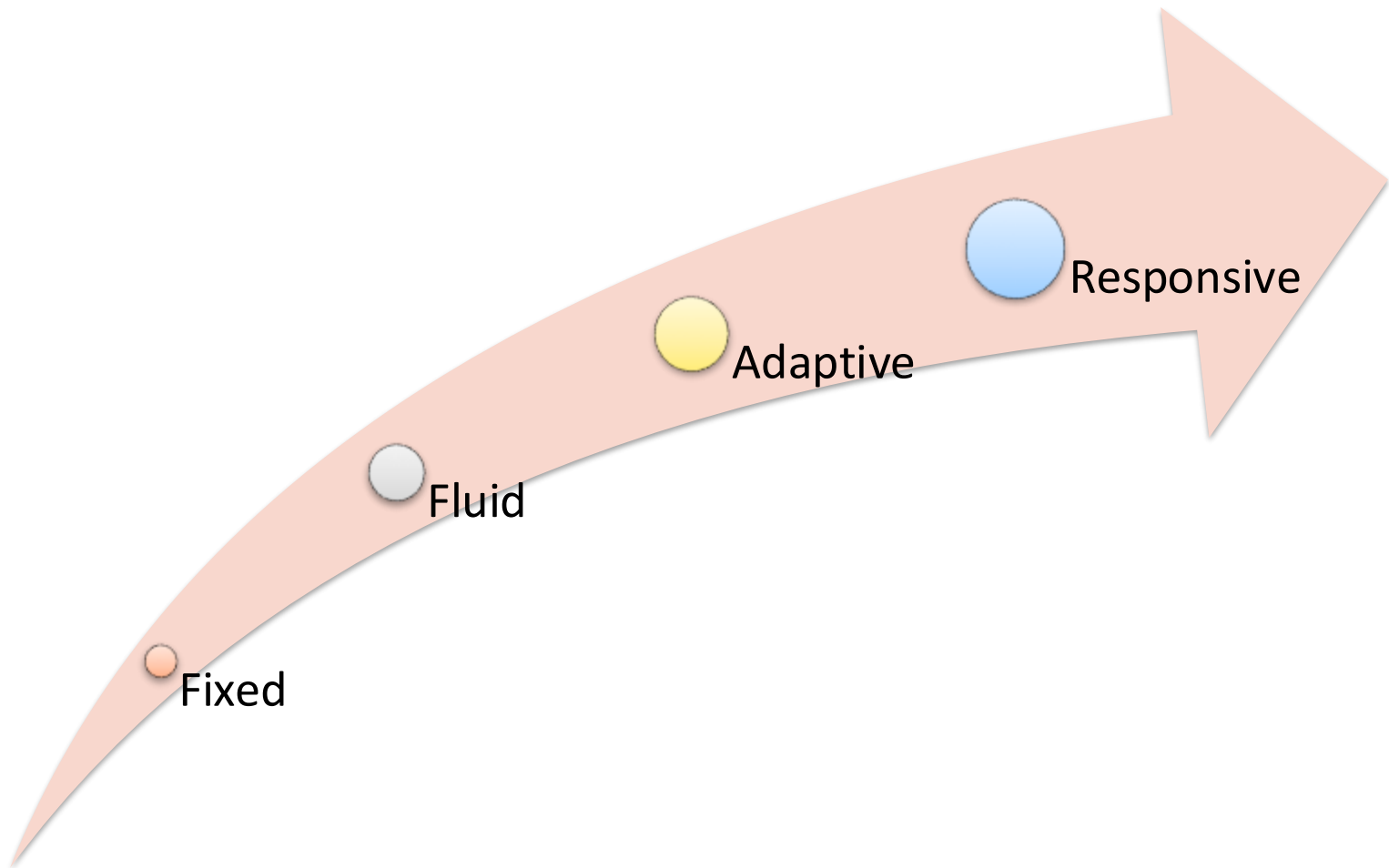
Prefer responsive as the default approach for most projects.

Use adaptive when you need highly optimized, different experiences per device.

Use fluid techniques inside responsive layouts to keep elements flexible.

Avoid pure fixed layouts unless you control the display environment.

Web designs in time



Summary

- Proper web design is essential to deliver quality of websites/apps
- There is a selection of many popular layouts, each with suggestions for correct usage
- Start selection by analysing the requirements
- If possible, use templates
- Preparing the styling consider different options to make CSS layouts

TILL NEXT LECTURE